Stones, Obstruction & Trauma

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 93	Management of Upper Urinary Tract Obstruction
Ch 94	Upper Urinary Tract Trauma
Ch 95	Urinary Lithiasis: Etiology, Epidemiology and Pathogenesis
Ch 96	Acute Care of Urinary Lithiasis
Ch 97	Surgical Management of Ureteral Calculi
Ch 98	Surgical Management of Upper Urinary Tract Calculi
Ch 99	Prevention of Urinary Stone Disease
Ch 100	Lower Urinary Tract Calculi
Ch 54	Management of Paediatric Kidney Stone Disease
Ch 125	Genital and Lower Urinary Tract Trauma
Ch 62	Paediatric Genitourinary Trauma
★	Putting it together

93 Management of Upper Urinary Tract Obstruction

1 · Ureteropelvic junction obstruction (UPJO): presentation

Why this matters: UPJO presents differently at each age.

- Trauma: see *Ch 94, Ch 125*; stones: see *Ch 95–100*.
- UPJO is usually congenital. **Neonates** show prenatal hydronephrosis on ultrasonography, with azotaemia if bilateral or solitary.
- **Older children and adults** have intermittent flank pain with vomiting, haematuria after minor trauma, urinary tract infection (UTI) and, rarely, hypertension.

2 · Diagnosis

Why this matters: a dilated kidney is not always obstructed, so prove impaired drainage before operating.

KEY POINT

Dilation is not obstruction. Anatomical imaging (ultrasonography, computed tomography [CT], retrograde) shows *where*; functional tests (renography, Whitaker) show *whether it matters*. Combine one of each.

Modality	Key point
Ultrasonography (first in neonates)	Large medial pelvis, rounded calyces joined by dilated infundibula (unlike multicystic kidney); ureter not seen
Non-contrast CT (acute pain)	Hydronephrosis to UPJ without stone; excludes radiolucent stone, tumour
Contrast CT	Shows **crossing vessels**; can substitute for renography
Diuretic renography	Differential function, drainage; best follow-up study
Retrograde pyelography	**At the time of repair** (avoids introducing infection)
Nephrostomy and antegrade study	When retrograde fails; tests recoverability; permits Whitaker

- P1 Radionuclide studies and diuretic renography in PUJ obstruction — *tracers, well-tempered protocol, T½ cut-offs, P40, O'Reilly curves*
- **Renography (Ch 93 values)**: hydrate, then give furosemide **0.3–0.5 mg/kg** at **20 min**. Mercaptoacetyltriglycine (MAG3) is preferred.
- **Obstruction is a half-time (T½) >40 min**. After successful pyeloplasty, drainage occurs even before furosemide.

ALGORITHM

Whitaker test

1. Place a nephrostomy and a bladder catheter; perfuse at **10 mL/min**.
2. If pressure stays ≤12–15 cm H_2O, there is no obstruction; if it exceeds **15–20 cm H_2O**, there is.
3. It is invasive, so use it only when other tests are non-diagnostic. Errors: pelvic compliance, position.

3 · Indications and choice of intervention

Why this matters: not every UPJO needs surgery, and risk factors predict which repair works.

- **Indications**: symptoms; impaired or falling function; stones or infection; rarely hypertension.
- **Neonates** are selectively observed; operate for progressive hydronephrosis or a **10%** fall in differential function. **Asymptomatic adults** with indeterminate results have serial renography.
- The **procedure of choice** is dismembered pyeloplasty, now robotic at most major centres; endopyelotomy fails more.
- Drain before repair only for infection, azotaemia (solitary or bilateral) or unrelenting pain.

Endopyelotomy subgroup (landmark series)	Success
Crossing vessel present vs absent	**42% vs 86%**
High- vs low-grade hydronephrosis	60% vs 81%
Both excluded	95% (≈ pyeloplasty)

- A crossing vessel is **prognostic, not a contraindication**, but it pushes primary UPJO towards pyeloplasty.
- **Secondary UPJO: cross over.** Failed endopyelotomy is treated by pyeloplasty; failed pyeloplasty by endourology first.
- **Nephrectomy** suits a thin shell with a normal contralateral kidney. Adults with < **15%** differential function are non-salvageable; children have no agreed cut-off.
- If uncertain, place a temporary stent or nephrostomy and re-scan. Nephrectomy also suits stones with chronic infection and multiple failed repairs.

4 · Endopyelotomy

Why this matters: recovery is quick, but success depends on careful selection.

- **Principle**: a full-thickness **lateral** incision into peripelvic fat heals over a stent. Lateral is chosen because it is **free of crossing vessels**.
- Pyeloplasty (≈ **95%**) wins because it makes a wide dependent funnel and reduces the pelvis; endourology wins only on recovery.
- **Contraindications**: segment > **2 cm**, active infection, uncorrected coagulopathy. Marked vessel entanglement on 3D CT calls for robotic repair.
- Urine must be sterile first; if obstruction prevents clearance, temporise with a stent or nephrostomy.

Percutaneous antegrade

- Best for **UPJO with pelvicalyceal stones**. Puncture the **mid-posterior or superolateral calyx**, not the lower pole, whose acute infundibular angle blocks passage. Upper-pole access risks hydropneumothorax.
- Cross the UPJ with a working and a safety wire before incising.
- Stent with a 14/7-Fr endopyelotomy stent (or 8–10 Fr), traditionally for **6 weeks**; 2 and 4 weeks are equal.
- **Clear stones before incising**, because fragments migrate periureterally and cause fibrosis. A pelvic stone may cause oedematous pseudo-UPJO, so clear it and reassess.

Retrograde ureteroscopic (standard)

- P1 Retrograde endopyelotomy — *indications, holmium lateral-incision steps, results (≈90% without crossing vessel >4 mm), complications, balloon/cautery-wire alternatives*

Not in the paper answer:

- Control bleeding with haemostatic laser settings or 10-min balloon tamponade.
- A narrow ureter is balloon-dilated, or stented and revisited in 5–10 days.

> ALGORITHM
> 1. General anaesthesia; working and safety wires; flexible ureteroscope via a sheath.
> 2. With a 200-µm holmium fibre, incise **laterally** into fat and into normal ureter.
> 3. Leave a stent and Foley; do a renogram 4 weeks after stent removal.

- **Percutaneous endopyeloplasty** is an endoscopic Heineke–Mikulicz repair; it fails in secondary UPJO.

Complications and follow-up

> ALGORITHM
> **Percutaneous bleeding**
> 1. Bed rest and transfusion.
> 2. **Do not irrigate** the nephrostomy, so clot tamponades.
> 3. If bleeding persists, do **selective embolisation** (low threshold).

- Renogram at 1 month after stent removal, at 6 months, then yearly for **2–3 years**. Pyeloplasty 85/75% vs endopyelotomy 63/41% at 3/10 years.
- After failure, get CT angiography; a significant crossing vessel needs **pyeloplasty, not repeat incision**.

5 · Pyeloplasty

Why this matters: dismembered pyeloplasty is the gold standard by any route.

> KEY POINT
> **Principles (open = minimally invasive)**: widely patent, watertight, tension-free anastomosis; funnel-shaped, **dependent** transition.

- P3 P1 Laparoscopic pyeloplasty; Management of failed pyeloplasty — *Anderson–Hynes steps, approaches, contraindications, stent 4–6 weeks, vascular hitch, late failures, redo options*

Not in the paper answer:

- Open approaches: flank extraperitoneal (default); anterior extraperitoneal for horseshoe or pelvic kidney; posterior lumbotomy if thin with no previous surgery.
- The transmesenteric route needs a visible dilated pelvis, which pre-stenting hides. Place the suction drain near, not on, the suture line.
- MCQ Trendelenburg 1886 made the first attempt; Kuster 1891 the first successful dismembered repair; **Anderson–Hynes 1949** is standard; Davis 1943 described intubated ureterotomy.
- Dismembered repair suits any insertion and allows pelvic reduction and **transposition over crossing vessels**. It is the only technique that excises the abnormal UPJ, but it suits long proximal strictures or a small intrarenal pelvis poorly.

ALGORITHM

1. Dissect the ureter with periureteral tissue; place a lateral marking stitch.
2. Excise the UPJ; spatulate **laterally**; bring the apex to the inferior pelvis, anterior to any crossing vessel.
3. Do a watertight absorbable anastomosis, with or without reduction; leave a stent and drain.

- Remove the **stent** at **4–6 weeks**. An added nephrostomy in uncomplicated adults means longer stay and more infection, so reserve it for secondary UPJO.
- Laparoscopic and robotic success is >**90–95%**, equal to open; the robot's advantage is ergonomic. "Strict" success definitions give rates about one-third lower.

ALGORITHM

1. Remove the Foley at 24–36 h, then the drain if output is negligible.
2. If output rises after Foley removal (reflux along the stent), **replace the Foley for 7 days**.
3. Remove the stent at 4–6 weeks; then renography.

6 · Other UPJ reconstructions

Why this matters: flaps or the calyx bridge gaps unsuited to dismemberment.

KEY POINT

Every UPJ repair asks: **what tissue bridges the gap while keeping blood supply?** Dismemberment excises disease but devascularises; flaps keep blood supply but carry disease forward; the calyx is used when the pelvis is too small.

Procedure	Indication	Note
Foley Y-V-plasty	**High insertion**	Not with crossing vessels or need for reduction
Culp–DeWeerd spiral flap	Large pelvis, long proximal narrowing	Longest flap; length:width ≤ **3:1**
Scardino–Prince vertical flap	Dependent UPJ, box-shaped pelvis	Shorter flap
Davis intubated ureterotomy	Long strictures, with spiral flap	Heals over stent (secondary intention)
Ureterocalicostomy	Small intrarenal pelvis, horseshoe, failed pyeloplasty	Ureter to lower-pole calyx

- **Y-V**: the V base lies on the dependent medial pelvis with its apex at the UPJ; the Y stem runs down the lateral ureter.
- **Flaps**: outline them longer than needed; trim the apex, never the base.
- **Intubated ureterotomy** uses a soft stent and nephrostomy. At 6 weeks: nephrostogram, stent out, clamp, then nephrostomy out.
- P2 Uretero-calycostomy — *resect lower-pole parenchyma (not nephrotomy), mucosa-to-mucosa, omental cover; cortex <5 mm predicts failure*

7 · Retrocaval ureter

Why this matters: the anomaly is venous, so the ureter is moved.

- P3 Retrocaval ureter — *venous anomaly, always right, fish-hook/S-shape, CT; transpose ureter anterior to IVC*

8 · Ureteral stricture — evaluation and endourology

Why this matters: a stricture may be cancer, and endoscopy works only for short, well-vascularised strictures.

- **Causes**: malignancy, stone, radiation, surgical ischaemia, periureteral fibrosis, endoscopy, infection (tuberculosis, schistosomiasis).
- **Exclude malignancy first** (goblet sign; metastases) with CT, with or without ureteroscopy and biopsy.
- Stricture follows ureteroscopy (URS) in ≈**3%** vs 1.5% after shock-wave lithotripsy (SWL); an impacted stone >2 months gives ≈1 in 4. Thulium gives more than holmium.
- Endourology needs ipsilateral function ≥**25%** (the algorithm uses 20%).
- **Stents** suit intrinsic strictures. Extrinsic compression needs tandem or metallic stents (changed yearly). Chronic stenting suits unfit or cancer patients.
- **Balloon dilation**: a 4-cm, 5–8-mm balloon is inflated until the **waist disappears**, for 10 min, then stented 2–4 weeks. Contraindications: infection, stricture >**2 cm**.
- Balloon results are best (≈85%) in short, recent (≤3 months), non-anastomotic, well-vascularised strictures; anastomotic ≈50%. Transplant strictures respond well.
- **Endoureterotomy** (retrograde holmium) is first line under 2 cm. Cut full thickness into **2–3 mm of normal ureter** each side, over a safety wire; larger stents (8–12 Fr) do better.

Level	Incision	Avoids
Distal/mid ureter	**Anteromedial**	Iliac vessels
Proximal ureter	**Lateral/posterolateral**	Great vessels

- **Complete obliteration** (high-risk patients only): use antegrade and retrograde scopes and **cut toward the light**, then stent 8–10 weeks.

ALGORITHM

1. If <2 cm and function >20%, then endoscopic incision.
2. If >2 cm and function >20%, then reconstruction.
3. If function <20%, then nephrectomy (children may still merit repair).
4. Transplant ureter: consider balloon. Follow with renography for 2 years.

KEY POINT

Each failure predictor (long, old, anastomotic, poorly vascularised) is a proxy for **ischaemic scar** that an incision cannot heal across.

9 · Ureteral stricture — reconstruction ladder

Why this matters: defect length and level choose the repair, and true length is often known only at surgery.

Define nature, site and length; exclude carcinoma; be ready for the next rung. Iatrogenic injury: see *Ch 94*.

Technique	Defect bridged	Level
Ureteroureterostomy	**2–3 cm**	Upper/mid
Ureteroneocystostomy	4–5 cm	Distal
Psoas hitch	**6–10 cm**	Below pelvic brim

Technique	Defect bridged	Level
Boari flap	**10–15 cm**	Lower/mid; spiral may reach pelvis
Renal descensus	5–8 cm	Upper; adds to any repair

- **Ureteroureterostomy** suits short defects only, because tension causes stricture. Preserve adventitia; spatulate 5–6 mm, 180° apart; place a double-J; cover with omentum.
- Remove the drain after 24–48 h of minimal output (check drain creatinine). Success is >90%.

ALGORITHM

Persistent leak

1. X-ray the stent position.
2. Pull the drain off the anastomosis and stop suction.
3. Place a Foley and give an anticholinergic; if it persists, place a nephrostomy.

- P3 P2 Boari flap; Trans Uretero Ureteric anastomoses — *contracted bladder contraindicates; recurrent stricture commonest; eGFR fell in 44%; TUU donor above IMA, contraindications, correct recipient reflux*
- In adults, refluxing and non-refluxing reimplants are equal; robotic matches open with shorter stay. Treat outlet obstruction first.
- **Renal descensus**: the kidney, in Gerota fascia, is rotated inferomedially and its lower pole fixed; the renal vein limits descent.
- **Ileal ureter** is isoperistaltic. Contraindications: creatinine >**2 mg/dL** (acidosis in ≈half), bladder dysfunction, inflammatory bowel disease, radiation enteritis.
- Use distal ileum ≥15 cm from the ileocaecal valve; segments >15 cm prevent reflux. Tapering is unproven. Dilation with acidosis means outlet obstruction; start endoscopic surveillance from year 3.
- **Buccal mucosa onlay** (robotic, omental wrap) suits dense scar. **Autotransplantation** suits an absent or poor contralateral kidney, or when other repairs are impossible.

KEY POINT

Ladder: bring the bladder up, the kidney down, borrow the other ureter, or substitute (ileum, buccal). Tension and ischaemia cause failure at every rung.

10 · Ureteroenteric anastomotic stricture

Why this matters: these strictures can appear years later and silently lose the kidney.

- Strictures occur in 4–9% of conduits, ≈5% after 10 years, so **upper-tract surveillance is lifelong**.
- **The cause is ischaemia** of the adventitial blood supply. The **left is commoner**, because it needs extra length, passes under the sigmoid mesentery and angulates at the inferior mesenteric artery (IMA).
- Risks: **non-refluxing anastomosis** (≈13% vs 2–3%), carcinoma in situ at the margin, obesity, American Society of Anesthesiologists (ASA) score >2, Clavien–Dindo 3–4 complications.
- Conduit dilation is often non-obstructive, so confirm with renography. An irregular mass with rapid progression suggests recurrent cancer.
- **Open revision is the gold standard** (≈91% vs 26% endoscopic); robotic is equal. Endourology is **antegrade**: balloon or holmium, then a flushable internal–external stent.
- Endoscopic success: <1 cm ≈50% vs >1 cm 6%; right ≈41% vs left 19%.

ALGORITHM

1. If <2 cm and function >20%: right side, endoscopic incision; **left side, formal repair** (sigmoid mesentery bleeding).
2. If >2 cm, then formal repair.
3. If function <20%, then nephrectomy or observation.

11 · Retroperitoneal fibrosis

Why this matters: retroperitoneal fibrosis (RPF) often presents with renal failure and may hide cancer.

- P3 P1 Retroperitoneal fibrosis — *idiopathic 70% vs secondary (drugs, lymphoma); plaque at L4–L5, medial deviation; PET for activity; deep biopsies; drain, then prednisone (80% respond), then bilateral ureterolysis and omental wrap*

ALGORITHM

1. If uraemic, decompress with a **nephrostomy** or stents and manage the **post-obstructive diuresis**.
2. Stop any culprit drug; biopsy.
3. Give steroids; if they fail, do ureterolysis.

- Steroids work while the lesion is cellular, so pain and erythrocyte sedimentation rate (ESR) fall within days. Second line: immunosuppressants, **tamoxifen**.
- Ureterolysis starts at the distal non-dilated ureter ("split and roll"); an omental wrap is most definitive.

94 Upper Urinary Tract Trauma

- P2 Renal trauma (renal parenchymal and vascular injuries, including grade IV): diagnosis and management — *mechanism, haematuria, AAST 2018, imaging indications, CT red flags, non-operative care, exploration, vascular control, renovascular injury, complications*

Not in the paper answer:

- Firearms cause about two-thirds of penetrating renal injuries.
- Polar injury: partial nephrectomy with omental cover. Also: high-risk grade IV, MiGUTS, post-repair hypertension, sport (below).

1 · Renal injury — mechanism and presentation

Why this matters: mechanism predicts the injury.

- The kidney is the commonest urological organ injured. Blunt trauma causes 80–90% (motor vehicle collision commonest); most are grade I–III and heal conservatively.
- **Deceleration** injures the fixation points (hilum and ureteropelvic junction, UPJ), causing artery thrombosis, vein disruption, and pedicle or UPJ avulsion.
- **Penetrating** injury causes more bleeding, nephrectomy and complications. High-velocity missiles cause blast and cavitation, so necrosis is delayed.
- Stabs at the anterior axillary line hit the hilum or pedicle; at the posterior axillary line, parenchyma. Mark gunshot entry and exit with metallic markers.
- Signs: flank ecchymosis and **ipsilateral rib 9–12 fractures** (strongest association).

2 · Haematuria

Why this matters: haematuria selects who is imaged, but its absence excludes nothing.

- Significant haematuria is **gross**, or **microhaematuria (>5 red blood cells per high-power field) with systolic blood pressure (SBP) <90 mm Hg**.
- Degree does not match severity; haematuria is absent in ~⅓ of blunt renovascular injuries and **~½ of UPJ injuries**.
- Test the **first aliquot**. Dipstick false positives: myoglobinuria, menses, povidone-iodine.

> **KEY POINT**
> Haematuria triggers imaging; it does not measure injury. The worst injuries (pedicle, UPJ) may not bleed into the urine, so mechanism carries equal weight.

3 · AAST kidney injury scale (2018)

Why this matters: the American Association for the Surgery of Trauma (AAST) grade, set on CT, predicts intervention and nephrectomy.

Grade	AAST 2018
I	Subcapsular haematoma and/or contusion, no laceration
II	Perirenal haematoma within Gerota's fascia; laceration **≤1 cm**, no extravasation
III	Laceration >1 cm, no collecting-system rupture; **any vascular injury or active bleeding contained within Gerota's**
IV	Collecting system laceration with extravasation; pelvis laceration or complete UPJ disruption; **segmental artery or vein injury**; active bleeding beyond Gerota's; segmental/complete infarction without active bleeding

Grade	AAST 2018
V	**Main renal artery or vein laceration** or hilar avulsion; devascularised kidney with active bleeding; shattered kidney

- Advance one grade for bilateral or multiple injuries, up to III; grade by the highest finding. MCQ Trap: main-vessel injury was grade IV in 1989 and is now V; segmental pseudoaneurysm or arteriovenous fistula (AVF) is now IV.

4 · Indications for imaging (AUA = EAU)

Why this matters: the American Urological Association (AUA) and European Association of Urology (EAU) agree.

- **Penetrating** trauma with **any haematuria** needs CT, if the patient is stable.
- **Blunt** trauma needs CT with rapid deceleration, gross haematuria, or microhaematuria with SBP <90 at any time.
- Blunt microhaematuria without shock or deceleration is **observed without imaging**.
- **Children** have larger kidneys, less fat and pliable ribs; catecholamines hold blood pressure until ~50% blood loss, so image liberally. UPJ obstruction lets minor trauma cause major injury.

5 · Imaging studies

Why this matters: without a delayed phase, urinary extravasation is missed.

- Gold standard: **contrast CT with arterial/nephrographic and delayed (~10 min) phases**. Vein injury is poorly seen; medial haematoma with normal arteries suggests it.
- Omit delays if kidneys are normal with no perinephric or pelvic fluid; if injury is suspected and none was obtained, repeat a urographic phase.
- Focused assessment with sonography for trauma (FAST) has poor specificity; it cannot separate haematoma from urine.

CT finding	Suggests
Medial haematoma	Vascular pedicle injury
Medial extravasation; no contrast in distal ureter	Renal pelvis/UPJ avulsion
Absent global enhancement (cortical rim sign)	Main renal artery occlusion
Contrast blush	Brisk bleeding; strongly predicts need for angioembolisation (AE)
≥2 of: rim distance >4 cm, medial laceration, blush	High-risk grade IV

- The **MiGUTS nomogram** predicts bleeding intervention from penetrating mechanism, blush, pararenal haematoma, associated injuries and shock; the largest contributor is **haematoma rim distance**.

ALGORITHM

One-shot intravenous pyelogram (IVP) (unstable, no CT): **2 mL/kg contrast → single film at 10 min**. It confirms a working contralateral kidney; alternatively palpate it, or loop the injured ureter and give IV methylene blue.

Perinephric collection	Hounsfield units (HU)	Feature
Urinoma	0–20	Contrast pooling on delayed phase
Haematoma	>30	—
Abscess	>20	Rim enhancement

6 · Nonoperative management

Why this matters: observation saves most kidneys, even high-grade.

- Stable, staged **grade I–IV and selected V** injuries are observed, whatever the mechanism. Success is >90% in blunt injury, even with extravasation. **Early mobilisation is safe**.
- **Failure triggers**: instability; transfusion of **2–3 units**; new pain or guarding; enlarging urinoma with fever. Most failures need only stent, nephrostomy or AE.
- **Penetrating** injury breaches Gerota's fascia, losing tamponade, so more fail; stabs do better than gunshots. A stable stab behind the posterior axillary line gets serial examination.
- Failure within 24 h is rare (~3%): higher grade, penetrating, nonrenal abdominal injury.
- **Reimaging** (AUA): CT at ~48 h for **grade IV–V deep lacerations** or deterioration, not for uncomplicated I–III.

7 · Urinary extravasation

Why this matters: most leaks heal, unless urine cannot reach the bladder.

- Parenchymal leaks **heal in 76–90%**. **Drain early** when urine cannot pass antegrade: pelvis or UPJ avulsion, ureteric clot, UPJ obstruction, parapelvic cyst.
- A symptomatic or enlarging urinoma gets a retrograde stent and catheter, then a nephrostomy or drain if needed.
- If retrograde stenting is impossible (urethral injury, transection, no lithotomy), place a nephrostomy, with or without an antegrade stent.
- Extravasation with >25% nonviable parenchyma gives more abscesses, so operate sooner.

> **KEY POINT**
> Parenchymal leaks heal because antegrade flow is preserved; anything stopping urine reaching the bladder converts an observable leak into one that must be drained.

8 · Intervention

Why this matters: needless exploration releases tamponade and costs kidneys.

Absolute indications	Relative (increasingly observed)
Instability, no or transient response	Extravasation with significant devascularisation
Expanding or pulsatile haematoma	Colon or pancreatic injury
Suspected pedicle avulsion; UPJ avulsion	Arterial thrombosis; parenchymal extravasation

- **AE** is superselective and raises salvage; repeat AE succeeds; it needs an experienced team, without delay. **Failure predictors**: instability, grade V, urinary extravasation, gross haematuria.
- At laparotomy, exploration is **not mandatory** for staged grade I–III. The only absolute indication is a **pulsatile or expanding haematoma**, because opening Gerota's fascia releases tamponade.

ALGORITHM

Vascular control first

1. Eviscerate the small bowel; incise peritoneum over the aorta above the inferior mesenteric artery (IMA), up to the ligament of Treitz.
2. Follow the aorta to the left renal vein; loop and retract it; loop both renal arteries.
3. Reflect the colon and open Gerota's fascia; occlude the vessels if bleeding.

- **Damage control** (hypothermia, acidosis, coagulopathy): pack, close temporarily, resuscitate in intensive care, return at 24–48 h; this may allow AE instead of nephrectomy.

ALGORITHM

1. Expose and gain vascular control; debride only devitalised tissue.
2. Suture-ligate vessels; close the collecting system **watertight**.
3. Approximate parenchyma over a bolster; cover with Gerota's fascia or **pedicled omentum**; drain.

- Colon or pancreatic injury no longer mandates nephrectomy.
- **Revascularise** only a solitary kidney or bilateral injury, with warm ischaemia **<120 min**; unilateral repair salvages only ~⅓.
- The **left renal vein may be ligated; the right cannot**.
- Hypertension follows thrombosis repair in ≥43%, so monitor blood pressure for a year. Endovascular stenting of an intimal flap needs anticoagulation, its drawback.

9 · Complications and follow-up

Why this matters: bleeding and hypertension can appear late.

- Acute kidney injury relates to age, transfusion and polytrauma, not nephrectomy. Delayed bleeding comes within 21 days; treat by AE.
- **Page kidney** (subcapsular compression): angiotensin-converting enzyme inhibitor or receptor blocker; if refractory, drainage or decortication. Goldblatt kidney is artery stenosis; post-traumatic AVF needs AE.
- EAU: no imaging for grade I–II; salvaged grade III–V get a dimercaptosuccinic acid (DMSA) scan.
- Return to sport at 2 weeks–3 months; bleeding risk peaks at **2–6 weeks**. A solitary kidney does not bar contact sport.

10 · Ureteral injury

Why this matters: ureteral injuries are often silent, and late diagnosis means fistula or kidney loss.

- P3 Ureteric injury: iatrogenic (gynaecological) and upper ureteric injury — *aetiology, risk sites, prevention, CT with delays/retrograde, damage control, repair by level (hitch 95–100%, Boari up to 12 cm, TUU, ileal ureter), follow-up*

Not in the paper answer:

- Delayed upper-ureter repair: appendix (right) or buccal graft. Also: pyeloperfusion, long-silk stump ligation (below).

AAST ureter grade	Description
I	Contusion/haematoma without devascularisation
II	Laceration **<50%** transection

AAST ureter grade	Description
III	Laceration ≥50% transection
IV	Complete transection, **<2 cm** devascularisation
V	Avulsion, >2 cm devascularisation

- Iatrogenic injury is commoner (distal); external is mostly gunshot (proximal). **Haematuria is absent in 25–45%**.
- The **distal ureter** is the commonest iatrogenic site. Minimally invasive hysterectomy is riskier than open; prolapse surgery is highest-risk; endometriosis raises risk.
- In ~10% of women the uterine artery mainly supplies the distal ureter, so devascularisation presents at 1 week–2 months (stenosis, urinoma, ureterovaginal fistula).
- **Ureteroscopy**: stricture ~3%, avulsion <1%. At a tear, stop basketing and stent. Holmium is safest; use ≤14-Fr sheaths in unstented ureters and a safety wire.
- Partial nephrectomy or ablation of lower-pole or hilar tumours causes late proximal stricture; pyeloperfusion protects. **Ureteroarterial fistula** (chronic stent with pelvic surgery or radiation) needs an endovascular covered stent.
- Preoperative stents aid **recognition**, not proven prevention. Intraureteral indocyanine green fluoresces for hours. Check efflux at cystoscopy after minimally invasive hysterectomy and prolapse surgery.
- Efflux aids: oral phenazopyridine (most cost-effective), IV indigo carmine (safe), fluorescein, dextrose irrigation. **Methylene blue** is a monoamine oxidase inhibitor (serotonin toxicity); avoid in pregnancy and glucose-6-phosphate dehydrogenase deficiency; it falsely lowers oximetry.

Modality	Role
CT urography, **10-min delays**	Best in a stable patient; trace both ureters
Retrograde ureterography	**Most sensitive**; allows stenting
Antegrade study	When retrograde fails
One-shot IVP	Often non-diagnostic; never rely on it

- CT signs: **no contrast distal to the injury**, medial extravasation (UPJ avulsion), delayed nephrogram, urinoma. ~11% are missed at laparotomy.

KEY POINT

Ureteral injury is diagnosed by absent contrast beyond it, so time the study for excretion, follow both ureters, and never treat a non-diagnostic study as negative.

ALGORITHM

1. If stable at laparotomy, repair now; if unstable, do damage control and repair later.
2. If contusion, stent, or excise and repair.
3. If found late or partial, stent; if stenting fails, place a nephrostomy and repair later.
4. If ureterovaginal fistula, try a stent first.

- **Damage control**: ligate the proximal stump with long silk, then nephrostomy once stable; or exteriorise a single-J stent or 5-Fr feeding tube. Open nephrostomy is slow.
- Principles: spare adventitia; debride to bleeding edges; make it **spatulated, tension-free, stented and watertight**; drain. **No bowel or autotransplantation acutely.** Close a

longitudinal laceration transversely.

- In trauma, reimplant **non-tunnelled**, because tunnelled reimplants stricture more.
- Psoas hitch beats distal ureteroureterostomy, because the distal stump's supply is tenuous. Suture to the psoas minor tendon, sparing genitofemoral and femoral nerves.

95 Urinary Lithiasis: Etiology, Epidemiology and Pathogenesis

1 · Overview and epidemiology

Why this matters: knowing who forms stones, and why, guides prevention.

- **Removal is surgical, but recurrence is metabolic**. Prophylaxis: see *Ch 99*.
- Lifetime prevalence is **1–15%**; in the United States ≈ **10%** and rising, partly from more computed tomography (CT).
- **Sex**: males outnumber females 2–3 : 1, but the gap is narrowing, with the steepest rise in adolescent females.
- **Oestrogen is protective** (raises renal calcium reabsorption, lowers bone resorption), hence a second female peak at menopause.
- **Race**: highest in White, lowest in Asian people. **Age**: peak in the 4th–6th decades; paediatric stones are rising 5–10% a year.
- **Climate**: hot, arid areas, the US Southeast "stone belt" and summer. Heat lowers **urine volume**; sunlight causes **vitamin D-mediated hypercalciuria**.
- **Occupation**: heat lowers urine volume and pH (uric acid stones); sedentary work and weightlessness cause hypercalciuria; restricted bathroom access lowers volume.
- **Obesity** (body mass index, BMI) is an independent risk, stronger in females. **Metabolic syndrome** risk rises with the number of traits; **diabetes** and stones predict each other.
- **Insulin resistance** unifies these: it lowers urine pH, causing uric acid stones, and hyperinsulinaemia causes hypercalciuria.
- Hypertension, myocardial infarction, stroke, chronic kidney disease (CKD) and end-stage renal disease are associated (causation unproven). Fluid intake is inversely related (proven by randomised controlled trial, RCT); water hardness matters little.

2 · Physicochemistry and stone formation

Why this matters: prevention lowers supersaturation or aids inhibition.

- P2 Stone formation (etiopathogenesis) and medical management of urolithiasis — *saturation states, heterogeneous nucleation, steps, Randall plaque vs ductal plugs, inhibitor list, matrix 2.5%*

Not in the paper answer:

- Free-particle theory (crystals trapped as plugs) vs fixed-particle theory (anchored on plaque); the vasa recta hypothesis of plaque.
- Brushite conversion after infection or lithotripsy (below); the role of matrix is undetermined.
- Stones form by **supersaturation, then nucleation, growth and aggregation, then retention**. The concentration product (CP) multiplies ion concentrations; the solubility product (Ksp) is CP at saturation; the formation product (Kf) is CP at spontaneous crystallisation.

State	CP	What happens
Undersaturated	< Ksp	No crystals; stones may dissolve
Metastable	Ksp–Kf (most urinary salts)	No homogeneous (spontaneous) nucleation; growth and heterogeneous nucleation possible; **inhibitors effective, so this is the zone of therapy**

State	CP	What happens
Unstable	> Kf	Spontaneous nucleation; inhibitors ineffective

- The **relative saturation ratio (RSR)** is CP ÷ Ksp. Below 1, stones dissolve; above 1, they grow. The EQUIL 2 program overestimates it; the JESS program matches direct measurement.
- Changing pH lowers the saturation of **phosphate, uric acid and cystine, but not oxalate**, because oxalic acid is strong (pK ≈ 4). **Calcium and oxalate contribute equally** to calcium oxalate saturation.
- Urine holds calcium oxalate at ≈ 4× its solubility in water; with inhibitors it precipitates only at **7–11×**.
- Nephron transit takes 5–7 min, too short for one crystal to block a tubule, so aggregation occludes.
- **Randall plaque** (MCQ 1937) is **calcium apatite**. It starts in the **thin-limb basement membrane**, spreads through the interstitium and erodes the papilla; calcium oxalate then grows on it. Osteopontin lies at the mineral–organic interface.

Stone former	Randall plaque	Ductal plugs
Idiopathic calcium oxalate	Present, as the anchor	Uncommon, mild
Intestinal bypass	Absent	Apatite plugs in inner medullary collecting ducts; injury, fibrosis
Brushite	Present	Apatite plugs; interstitial fibrosis
Distal renal tubular acidosis (RTA), hyperparathyroidism, cystinuria	—	Present

KEY POINT

Plaque vs plug: a plaque is an idiopathic interstitial apatite anchor; a plug is intratubular obstruction with injury, caused by secondary systemic disease. **Ductal plugs should prompt a search for a secondary cause.** Infection or shock-wave lithotripsy (SWL) may convert a calcium oxalate former into a brushite former.

3 · Inhibitors and promoters

Why this matters: urine is supersaturated; inhibitors prevent stones.

- Inhibitors raise the supersaturation needed for nucleation. **Uric acid and cystine have no inhibitor**; only pH and concentration matter. Promoters: low volume and citrate; high calcium, oxalate, phosphate, uric acid.

Inhibitor	Action	Key feature
Citrate	Complexes calcium; inhibits precipitation, agglomeration and urate-induced nucleation	Most important small-molecule inhibitor
Magnesium	Complexes oxalate	Synergy with citrate, phytate
Glycosaminoglycans	Bind crystal surface	Lower in stone formers
Nephrocalcin	Inhibits calcium oxalate monohydrate (COM) nucleation, growth, aggregation	Formers excrete isoforms lacking γ-carboxyglutamic acid

Inhibitor	Action	Key feature
Tamm-Horsfall protein	Inhibits aggregation, not growth	Most abundant urinary protein, made in the thick ascending limb (TAL); **a promoter in acid urine**
Osteopontin	Inhibits nucleation, growth, aggregation, epithelial binding	Inducible

- **Matrix** is ≈ **2.5%** of stone weight and contains matrix substance A (found in formers only).

4 · Mineral metabolism

Why this matters: normal handling explains each abnormality.

- **Calcium**: 30–40% is absorbed in the small bowel, transcellularly (calcitriol-dependent, saturable) at low intake and paracellularly at high intake.
- Luminal oxalate, phosphate and fatty acids bind calcium. So **oxalate-rich food lowers calcium absorption, and luminal calcium lowers oxalate absorption**.
- **Calcitriol** is made by 1α-hydroxylation in the proximal tubule, stimulated by **parathyroid hormone (PTH) and hypophosphataemia**. It is the most potent stimulator of gut calcium absorption.
- **PTH** causes bone resorption; in the kidney it raises calcium reabsorption, lowers phosphate reabsorption and raises calcitriol. It has **no direct gut effect**.

Segment	Filtered calcium reabsorbed	Route
Proximal tubule	**60–65%**	Paracellular; claudin-2
TAL	25–30%	Paracellular, lumen-positive; claudin-16/19; the calcium-sensing receptor (CaSR) acts through claudin-14 to block it
Distal convoluted tubule (DCT)	8–10% (fine control)	Transcellular: entry by TRPV5 channel, then carried by calbindin-D28k

- Only 1–3% of filtered calcium is excreted.
- **Oxalate**: 6–14% is absorbed, unregulated, and excreted almost entirely in urine; half is dietary, half hepatic (from glyoxylate).
- **Ascorbic acid** is the largest endogenous source (up to 40%); hydroxyproline ≈ 15%. *Oxalobacter formigenes* degrades oxalate and stimulates colonic secretion.

5 · Classification

Why this matters: the urinary abnormality, not the stone, decides treatment.

Stone	Frequency
Calcium oxalate (pure or with apatite)	**60%**
Hydroxyapatite	20%
Brushite	2%
Uric acid	≈ 7% (8–10% US, rising)
Struvite	≈ 7% (falling)
Cystine	1–3% (up to 10% children)
Triamterene, silica, 2,8-dihydroxyadenine (2,8-DHA)	< 1% each

Condition	Defect	% of formers
Absorptive hypercalciuria (incl. phosphate leak)	Increased gut calcium absorption	20–40
Renal hypercalciuria	Renal calcium leak	5–8
Resorptive hypercalciuria	Primary hyperparathyroidism (PHPT)	3–5
Hyperuricosuric calcium	Purine excess	10–40
Hypocitraturic calcium	Idiopathic, diarrhoea, distal RTA, thiazide	10–50
Hyperoxaluric calcium	Primary, enteric, dietary	2–15
Hypomagnesiuric calcium	Reduced gut magnesium absorption	5–10
Idiopathic low urine pH	Low pH	15–30
Low urine volume	Inadequate intake	10–50
Infection; cystinuria	Urease; cystine transport	1–5; < 1
No abnormality	—	< 3

KEY POINT

Classify by the **urinary abnormality, not the stone**: composition starts the work-up. Fewer than 3% have no abnormality. Work-up: see *Ch 99*.

6 · Hypercalciuria

Why this matters: fasting urine and PTH reveal the source.

- It is the commonest abnormality (35–65%); it raises calcium salt saturation and lowers inhibitor activity.
- **Definition**: > **200 mg/day** after 1 week on a 400 mg calcium, 100 mEq sodium diet; or > 4 mg/kg/day; or > 7 mmol/day (men) or 6 mmol/day (women).
- The **Pak classification** (gut, kidney, bone) is an exam framework, not a therapy guide.

Feature	Absorptive	Renal	Resorptive
Defect	Increased gut absorption	Renal leak	PHPT
Serum calcium	Normal	Normal	**High**
PTH	**Suppressed**/normal	**High** (secondary)	High
Fasting urinary calcium	**Normal** (< 0.11 mg/dL glomerular filtrate)	**High**	High
Serum phosphate	Normal (low in phosphate leak)	Normal	Low
Sequence	Hyperabsorption suppresses PTH and raises the filtered load	Renal loss raises PTH, which raises calcitriol	PTH drives bone resorption and calcitriol, causing hypercalcaemia

- Absorptive **type I** persists on calcium restriction; **type II** normalises.
- **CYP24A1** loss stops calcitriol inactivation, causing infantile hypercalcaemia, and adult hypercalciuria and nephrocalcinosis, worse with vitamin D.
- In **renal phosphate leak** (SLC34A1/A3), hypophosphataemia raises calcitriol, which increases gut calcium absorption.

Disorder	Gene	Features
Dent disease (X-linked)	ClC-5	Hypercalciuria, **low-molecular-weight proteinuria**, nephrocalcinosis, renal failure
Familial hypomagnesaemia with hypercalciuria and nephrocalcinosis (FHHNC)	Claudin-16, -19	TAL magnesium/calcium loss
Bartter	NKCC2, ROMK, ClC-Kb, barttin; type V activating CaSR	Salt wasting, hypokalaemic alkalosis, hypercalciuria

KEY POINT

Fasting urine and PTH separate gut from kidney: in renal leak, fasting calcium and PTH are high; in absorptive hypercalciuria, fasting calcium is normal and PTH suppressed.

- **PHPT** causes ≈ 5% of stones, and stones occur in < 5% of PHPT. Suspect it if serum calcium is > **10.1 mg/dL**; repeat with intact PTH, and use ionised calcium for normocalcaemic PHPT.
- A **thiazide challenge** unmasks PHPT (hypercalcaemia without PTH suppression). PTH also causes phosphaturia and mild hyperchloraemic acidosis. Others: thyrotoxicosis, vitamin D toxicity.

Cause	Mechanism	PTH	Clue
PHPT	Autonomous PTH	High/inappropriately normal	Commonest in outpatients
Sarcoidosis	Macrophage 1α-hydroxylase	Suppressed	Corrects with **corticosteroids**
Malignancy	PTH-related protein (PTHrP), osteolysis (RANKL)	Suppressed	Commonest in inpatients
Glucocorticoid excess	Increased bone resorption	Raised	Cushing syndrome

7 · Hyperoxaluria

Why this matters: bowel disease and diet are treatable causes.

- **Definition**: > **40 mg/day**. Types: primary, enteric, dietary and idiopathic.
- P2 Primary Hyperoxaluria — *PH1–3 enzymes, markers, course; oxalate ≥100 mg/day, oxalosis; pyridoxine (Gly170Arg); lumasiran; liver–kidney transplant in PH1*
- Exclude primary hyperoxaluria (PH) in relatives before living-related donation.

ALGORITHM

Enteric hyperoxaluria

1. Fat malabsorption leaves fatty acids in the gut lumen.
2. Fatty acids saponify calcium and magnesium, so less calcium binds oxalate and more oxalate is free.
3. Free oxalate undergoes **colonic hyperabsorption**, helped by bile salts that increase colonic permeability.

- **Enteric** is the commonest acquired form. Causes: small-bowel resection or Crohn disease, jejunoileal bypass, cystic fibrosis, and malabsorptive bariatric surgery (Roux-en-Y, duodenal

switch; not band or sleeve).

- It **needs an intact colon**; with an ileostomy, oxalate is normal but uric acid or ammonium urate stones form.
- Diarrhoea also causes hypokalaemia, hypocitraturia, hypomagnesiuria, low urine pH.
- **Dietary**: nuts, chocolate, tea, spinach, rhubarb. **Severe calcium restriction is counter-productive**, as less calcium binds gut oxalate. Vitamin C raises oxalate.
- Formers carry less *O. formigenes*; antibiotics deplete it. Mild idiopathic hyperoxaluria may matter as much as hypercalciuria (Arabian Peninsula).

8 · Hyperuricosuria, hypocitraturia, hypomagnesiuria

Why this matters: each is a common, treatable driver of calcium stones.

- **Hyperuricosuria** is > **600 mg/day** (diet, gout, myeloproliferative disease, hypoxanthine-guanine phosphoribosyltransferase [HGPRT] deficiency).
- Urate promotes calcium oxalate stones by **adsorbing inhibitors, "salting out" and epitaxy** (crystals growing on urate). Allopurinol cut recurrence (trial below).
- **Hypocitraturia** is < **320 mg/day**; **acid–base status is the main determinant**. Causes: distal RTA, chronic diarrhoea, animal protein, thiazides (through hypokalaemia), angiotensin-converting enzyme (ACE) inhibitors.

> **KEY POINT**
> Citrate excretion reads the proximal-tubular **intracellular pH**: acidosis, hypokalaemia or an acid diet make the tubule reclaim citrate. Hence potassium citrate is the treatment (see *Ch 99*).

- Hypomagnesiuria: **two RCTs of magnesium showed no benefit**. **Low urine pH** (< 5.5, formerly "gouty diathesis") causes uric acid and calcium oxalate stones.

9 · Renal tubular acidosis

Why this matters: distal RTA causes calcium phosphate stones.

- P2 Renal tubular acidosis — *type 1/2/4 comparison (urine pH, K^+, citrate, stones up to 70%), incomplete dRTA, ammonium chloride load, topiramate/acetazolamide*

Not in the paper answer:

- Hypercalciuria in distal RTA comes from acidosis demineralising bone; stones lead to the diagnosis in up to half. Genetics (below).

Feature	Type 1 (distal)	Type 2 (proximal)	Type 4
Defect	Distal H^+ secretion	Proximal HCO_3^- reabsorption	Aldosterone deficiency/resistance
Serum K^+	Low	Low	**High**
Urine pH	**Always > 6.0**	< 5.5 at steady state	Low
Urinary citrate	**Profoundly low**	Near normal	—
Stones	**Up to 70%**: calcium phosphate, nephrocalcinosis	Uncommon	Uncommon
Associations	Sjögren, lupus, obstruction; genetic	Fanconi	Diabetic nephropathy

- **Incomplete distal RTA**: electrolytes are normal, but urine pH cannot fall below 5.5 after an ammonium chloride load.
- In distal RTA, **profound hypocitraturia** is probably the most important stone factor.
- Genetics: SLC4A1 (anion exchanger 1, dominant); ATP6V1B1/ATP6V0A4 (recessive, with **sensorineural hearing loss**). Type 2 is self-limiting and spares citrate.

KEY POINT

Stone risk in RTA follows citrate and pH: only distal RTA combines fixed alkaline urine with profound hypocitraturia and hypercalciuria, the recipe for **calcium phosphate** stones.

10 · Uric acid stones

Why this matters: low urine pH causes them, so they dissolve medically.

- P2 Uric acid calculi: etiopathogenesis, medical management and allopurinol — *no uricase, pKa 5.35, three determinants (low pH dominant), insulin resistance/ammoniagenesis, idiopathic vs hyperuricosuric calcium*

Not in the paper answer:

- Uric acid forms about one-third of stones in type 2 diabetes.

Determinant	Causes	Importance
Low urine pH	Insulin resistance/obesity, animal protein, diarrhoea, gout	**Single most important**
Low volume	Diarrhoea, heat	Concentrates uric acid
Hyperuricosuria	Purine diet, gout, myeloproliferative disease, uricosurics	Least often the driver

- Humans lack hepatic **uricase**, so uric acid is not converted to allantoin (10–100× more soluble). Sodium urate is ≈ 20× more soluble than free acid.
- The pKa of uric acid is **5.35**. Undissociated uric acid dissolves only to ≈ 96–100 mg/L, but at pH 6.5 > 1200 mg/L stays soluble.
- Stones are radiolucent on kidney–ureter–bladder (KUB) film but visible on non-contrast CT (NCCT). BMI ≥ 27 predicts urine pH ≤ 5.5.

ALGORITHM

Why the urine is acid in insulin resistance

1. Insulin resistance and lipotoxicity make the tubule burn fatty acids, not glutamine, and lower sodium–hydrogen exchanger 3 (NHE3) activity.
2. Proximal ammoniagenesis falls, so H^+ is excreted as titratable acid.
3. Urine pH stays **persistently low**, uric acid stays undissociated, and stones form.

KEY POINT

Uric acid stones are a disease of **urine pH, not uric acid**. No inhibitor exists, so raising pH above the pKa removes the driver; †alkalinisation can dissolve stones (see *Ch 99*).

11 · Cystine and struvite stones

Why this matters: both recur aggressively without lifelong prevention or complete clearance.

- P1 P4 Cystinuria; Struvite stones — *SLC3A1/SLC7A9 types, solubility by pH, hexagonal crystals, CKD; urease cascade, biofilm, Proteus/Ureaplasma, discordant cultures*

Not in the paper answer:

- Cystinuria causes no nutritional deficiency, because amino acids are absorbed as oligopeptides.
- Most staghorns are calcium phosphate, not infection stones; voided urine culture is unreliable.

- **Cystinuria genes**: rBAT/SLC3A1 (chromosome 2p, type A), $b^{0,+}$AT/SLC7A9 (19q, type B), or both (type AB). Ornithine, lysine and arginine are also lost but are soluble.
- Incidence 1 in 1000–17,000; first stone at ≈ 12 years; males more severe. Coexisting hypocitraturia is commonest; ~¼ of adults have estimated GFR < 60.
- †Crystals are hexagonal; the cyanide–nitroprusside test screens for cystine.
- It is autosomal recessive; only cystine stones form. With **no inhibitor**, solubility is useful only at alkaline pH (≈ 1000 mg/L at pH 9).
- **Struvite**: ***Proteus mirabilis*** is commonest; also *S. aureus* and **Ureaplasma** (easily missed). *E. coli* is rarely urease-positive, so bladder and stone cultures are discordant.
- Struvite forms 5–15% of stones, female : male 2 : 1. Risks: old age, diabetes, stasis, urinary diversion, spinal cord injury.
- Urease is absent from sterile urine. Bacteria are encased within the stone, so clearance must be complete. **Less than half of staghorns are infection stones**. †Struvite crystals are "coffin-lid" shaped.

ALGORITHM

Urease cascade

1. Urease splits urea into ammonia (NH_3) and CO_2.
2. Ammonia forms NH_4^+ and OH^-, raising urine pH to 7.2–8.0.
3. With magnesium, phosphate forms **struvite**; with calcium, carbonate apatite.

12 · Miscellaneous and drug-related stones

Why this matters: rare radiolucent stones mimic uric acid; drugs are reversible causes.

	Xanthine	2,8-DHA
Enzyme deficient	Xanthine dehydrogenase (XDH)	Adenine phosphoribosyltransferase (APRT)
Radiology	Both radiolucent, so mistaken for uric acid	
Treatment	—	Allopurinol or febuxostat (more effective)

- **Ammonium acid urate**: radiolucent; follows diarrhoea, inflammatory bowel disease, ileostomy, **laxative abuse**, urinary tract infection (UTI); endemic paediatric bladder stones.

Route	Drug	Mechanism / stone
Direct	**Indinavir**	Insoluble at physiological pH; stones 4–13%; †radiolucent even on CT
Direct	Triamterene; guaifenesin/ephedrine; silicates	Drug stones
Indirect	Acetazolamide, **topiramate**, zonisamide	Mimic distal RTA, causing calcium phosphate stones
Indirect	Loop diuretics	Hypercalciuria (up to 64% of low-birth-weight infants on furosemide)
Indirect	Steroids, vitamin D, thiazides, laxatives, cytotoxics	Hypercalciuria, hypocitraturia, urate

13 · Anatomic predisposition and pregnancy

Why this matters: anatomical formers still need metabolic work-up.

Anomaly	Stones	Metabolic findings
Ureteropelvic junction obstruction (UPJO)	≈ 20%; children ~70× risk	Common (hypercalciuria first); recurrence after repair, often contralateral
Horseshoe kidney	≈ 36% adults	All tested abnormal; **hypocitraturia**; calcium stones ≈ 89%
Calyceal diverticulum	Up to 40%	Abnormal in half; UTI up to 40%
Medullary sponge kidney	Frequent, with nephrocalcinosis	Hypercalciuria, hypocitraturia; distal RTA on acid loading

KEY POINT

Anatomy sets the stage but metabolism writes the stone: anatomical stone formers still need **metabolic evaluation**, and pyeloplasty alone does not prevent recurrence (see *Ch 99*).

- **Pregnancy**: stones occur in 1 in 204–3000, in the 2nd–3rd trimester. Physiological hydronephrosis affects up to 90%, **right more than left**, resolving 4–6 weeks post partum.
- Raised glomerular filtration rate (GFR) and placental calcitriol promote stones; raised citrate and magnesium inhibit them. The **net risk is unchanged**, but stones skew to **calcium phosphate**.

14 · Landmark trials

Why this matters: they underpin fluid and allopurinol advice.

Trial	Design	Result	Take-home
Borghi water trial	RCT, high fluid vs no advice, first-time idiopathic calcium formers	Recurrence 12% vs 27%	Urine volume is the one proven universal modifier (see *Ch 99*)
Ettinger allopurinol trial	RCT, allopurinol vs placebo, hyperuricosuric normocalciuric calcium oxalate formers	> 2-fold fewer recurrences	Allopurinol for hyperuricosuric calcium oxalate stones (see *Ch 99*)

96 Acute Care of Urinary Lithiasis

1 · Presentation and diagnosis

Confirm the stone and spot infection early: fever with obstruction is an emergency.

- **Renal colic** is severe flank-to-groin pain; the patient writhes.
- A ureterovesical junction (UVJ) stone causes penile-tip pain and frequency.
- Haematuria occurs in ~**90%**, so it is absent in >10%.
- **Fever means an infected obstructed system**, needing urgent intervention.
- Send bloods, urinalysis and a **pregnancy test** in women of reproductive age.
- Urine >50 white cells (WBC)/hpf or nitrites suggest infection; raised blood WBC may be stress.

Modality	Role	Limitation
Non-contrast computed tomography (NCCT)	Gold standard (sens 97%, spec 95%); shows alternative diagnoses	8–16 mSv; misses matrix and indinavir stones
Low-dose NCCT (≤**3 mSv**)	Equal for stones ≥3 mm	Weak <3 mm and in obesity
Ultrasound (US)	First line in pregnancy, children, known formers	Up to 40% still need CT
Plain X-ray (KUB)	**Avoid** for acute diagnosis	Lowest accuracy
Magnetic resonance imaging (MRI)	Second line in pregnancy and children	Poor stone detection

- The STONE score failed validation as an NCCT substitute.

> **KEY POINT**
> Unknown diagnosis: NCCT answers everything at once. Known stone: US showing hydronephrosis suffices.

2 · Pain management

The first-line drug treats the cause of the pain.

- Obstruction stretches the system and releases prostaglandins, causing diuresis, raised intrarenal pressure and spasm. **Non-steroidal anti-inflammatory drugs (NSAIDs)** block this, matching opioids with less vomiting and rescue.

Agent	Place	Note
Ketorolac 30 mg	**First line** (EAU, NICE)	Avoid in renal failure, GI bleed, pregnancy
IV paracetamol	Second line	NSAID barred or insufficient
Opioids	Intractable pain	**Continue the NSAID**: morphine plus ketorolac beats either
Antispasmodics; forced fluids	**Ineffective**	Fluids only for vomiting

- Opioid dependency rises with each day beyond day 3.

3 · Medical expulsive therapy (MET)

MET helps selected stones pass.

- P2 Medical expulsive therapy (MET) in urinary calculi — *α1 mechanism, tamsulosin/silodosin, passage-by-width table, SUSPEND/STONE, benefit in distal 5–10 mm, intervene after 4–6 weeks*

Not in the paper answer:

- A 6 mm wide stone passes in ≈ 33%; a mid-ureteric stone in ≈ 60%.
- Measure **transverse width**, not length.
- Trial of passage, with or without MET, lasts **4–6 weeks**.

4 · Urgent intervention

These criteria decide who is drained now.

- **American Urological Association (AUA) indications**: obstruction with infection, bilateral obstruction, solitary kidney, immunosuppression, renal deterioration, refractory pain or vomiting.
- Uninfected obstruction harms only if prolonged (>~2 weeks).
- **Stent and nephrostomy (PCN)** are equal for sepsis; stents cause dysuria and haematuria, PCN a longer stay.
- **Stent failure** is predicted by P-CUT ≥**7.5 mm** (strongest), stone ≥9.5 mm and multiple stones; prefer PCN then.
- P-CUT is ureteric diameter at the stone minus stone width.
- Acute shock wave lithotripsy (SWL; proximal, <24 h) or ureteroscopy (URS; distal <10 mm) is done **only without infection** *(Ch 97)*.

ALGORITHM

1. Give analgesia; get NCCT.
2. If infection or an AUA criterion, then drain only (stent/PCN); treat the stone later.
3. If not and stone <10 mm, then trial with or without MET for 4–6 weeks.
4. If it fails, then URS or SWL.

5 · Special populations

Pregnancy, age and transplant change imaging and drainage.

- P2 Stones (urolithiasis) in pregnancy — *US, then MRI, then low-dose CT; conservative first, stent/PCN encrustation, URS any trimester*

Not in the paper answer:

- Hydronephrosis below the iliac vessels suggests stone.
- Resistive index difference ≥**0.06** means obstruction.
- ~30% need intervention; change stent/PCN every **4–6 weeks**.
- Early gestation favours PCN; preterm delivery is highest with PCN. URS is safe in any trimester (ACOG 2019).

Population	Pitfall	Key point
Elderly	Atypical pain; first stone after 60 unusual	Always CT to exclude aortic/cardiac mimics; drain early
Transplant kidney	Denervated, so **painless**	Fever or renal failure; endoscopic, else percutaneous

6 · Landmark trials

Trial	Design	Result	Take-home
SUSPEND	RCT, tamsulosin vs placebo, any ureteral stone, n = 1136	Intervention at 4 weeks 19% vs 20%	Unstratified; does not abolish MET for distal 5–10 mm

97 Surgical Management of Ureteral Calculi

- P2 Management of upper ureteric stones — *passage rates, EAU choice by size, SWL vs URS table, SWL failure factors, flexible URS, antegrade and laparoscopic routes, complication rates*

Not in the paper answer:

- Shock wave lithotripsy (SWL) **within 48 h** of an acute obstructing stone beats delayed SWL.
- Little is gained beyond 2 SWL sessions; salvage ureteroscopy after failed SWL is not harder; **forced diuresis is useless**.

1 · Background and indications

Decide who needs treatment, and when.

- The dangers are **renal damage and sepsis**.
- Non-contrast CT (NCCT) is the gold standard; low-dose is equal. Ultrasound (US) misses mid-ureteric stones.
- After NCCT localisation, US with a plain film (KUB) suits follow-up; radiolucent stones are invisible.
- Medical expulsive therapy (MET) is controversial after SUSPEND *(Ch 96)*; intervene after **4–6 weeks**.
- The ladder is SWL, then ureteroscopy (URS), then percutaneous nephrolithotomy (PCNL) with antegrade URS.

Tier	Trigger
Elective	Failed trial of passage
Urgent	Unrelenting pain; obstruction-related renal dysfunction
Extremely urgent	Obstructive pyelonephritis or sepsis: **drain only**, never manipulate the stone

- Work-up: blood count, creatinine and **urine culture**; exclude coagulopathy and untreated urinary infection (UTI). A preoperative KUB confirms the stone is radio-opaque.

Modality	Antiplatelet/anticoagulant	Bridging
PCNL	Stop if feasible	Short-acting heparin if high risk
SWL	Stop if feasible	Not recommended; choose another modality
Laser URS	**Need not stop**	None; more haematuria and stent symptoms

> **KEY POINT**
> Why only URS tolerates anticoagulation: SWL and PCNL injure parenchyma where bleeding is unseen and uncontrollable; laser URS works inside the lumen under vision.

2 · Choosing treatment

Size and impaction tip the choice.

- Upper ureter ≤2 cm: stone-free rate (SFR) at 3 months is similar, but SWL needs more retreatment.
- Distal ureter: SFR is similar; SWL needs more sessions but has fewer complications.
- **Impaction** (thick wall around the stone on NCCT) defeats SWL, so treat endoscopically.
- A large hard proximal stone needs PCNL with antegrade URS; if unfit, staged URS.

Modality	Size guidance
SWL	Historical limit 2 cm; SFR falls above **10–15 mm**
URS	No definite limit
PCNL with or without antegrade URS	>15–20 mm, large hard proximal

KEY POINT

Constant trade-off: SWL means fewer complications but more sessions; URS means one session with slightly more morbidity.

3 · Ureteroscopes

Match the scope to stone site and ureter calibre.

	Semirigid	Fibreoptic flexible	Digital flexible	Single-use digital
Image	Stable unless bent	Degrades as fibres break	No degradation	New each case
Calibre	Tapered	Smallest tip	~8.5 Fr	7.7 Fr
Cost	Cheap	Repair ~every **15** cases		Low volume

- Semirigid scopes suit the distal and mid ureter; torque at the brim damages them.
- Flexible scopes suit proximal stones, following any migration.
- Digital scopes are **not more durable**; keep a fibreoptic scope for a narrow ureter.
- Raised intrarenal pressure (IRP) causes fever and sepsis.

4 · Access

Most injuries happen at access, so be safe before fast.

- P3 Accessories for RIRS; Ureteric stents and ureteroscopy with or without stenting — *guidewire types, 12/14 Fr sheath, safety wire (Segura), pre-stent 2–4 weeks, stent indications, Denstedt stentless trial*

Not in the paper answer:

- Second look after pre-stenting: ≥14 days for best SFR, ≤4 weeks for lower sepsis.
- ~1 in 10 ureters cannot be accessed safely.
- Reserve the sheath for stones **above the iliac vessels**.
- **α-Blockers** ~1 week pre-URS give easier access, higher SFR and less stent pain.
- Consent covers access failure, further procedures, perforation, avulsion, sepsis, stricture and nephrectomy.
- A **safety wire** is advised, because complexity is unpredictable.
- For a difficult ureter, place a stent and return after **2–4 weeks**.

ALGORITHM

1. If the wire will not pass, place an open-ended catheter 1–2 mm short of the stone.
2. Do a retrograde pyelogram.
3. Use contrast hydrodissection with a hydrophilic wire.
4. Advance the catheter past the stone, then exchange for a stiff wire.

- If the catheter abuts the stone, the wire exits at 90°, causing **perforation**.
- If the ureter is inaccessible, **stent and stage**.
- If the scope will not enter the orifice, use **railroading** between two wires, a dual-lumen catheter, scope rotation or an obturator.

Dilation method	Best for
Indwelling stent	Whole ureter narrow; gentlest, but commits to a second procedure
Serial/coaxial dilators	Long narrow segments
Balloon	Focal stenosis; inflate until the waist resolves, hold 1–2 min
Semirigid scope	Distal ureter

- **Pre-stenting is not needed** for ureteral stones; it helps only renal stones.
- The access **sheath** helps with large burden, dusting and high power, lowering IRP and heat.
- 14/16 Fr sheaths reduce ureteral blood flow, though stricture is rare.

5 · Lithotripsy

Fragment safely, and keep heat down.

- **Basketing**: keep the stone **5–10 mm from the tip**, **never basket blind**, and when in doubt **fragment first**.
- Entrapped basket: push it to a wider segment, disassemble it, pass the scope alongside and laser the stone; else cut the basket.

Source	Key points
Electrohydraulic	High ureteral trauma; abandoned
Pneumatic	~12 Hz; **rigid/semirigid only**; best below brim; retropulsion
Holmium laser (Ho:YAG)	Short pulse fragments; long or modulated pulse lowers retropulsion; dusting is low energy, high frequency
Thulium fibre laser (TFL)	Low peak power and long pulse give fine dust; minimal retropulsion; >40 Hz misses

- P4 Thalium laser in urology — *TFL physics, settings (0.6 J × 6 Hz; dust 0.2 J × 40 Hz), "paint, do not drill", RCTs vs holmium*

Not in the paper answer:

- Impacted stone: target the centre first, so the sides fall away from the wall.

Setting	Ho:YAG	TFL
Fragmentation	0.6–1.0 J × 6–10 Hz	0.6 J × 6 Hz short pulse (3.6 W)
Dusting	0.2–0.3 J × 40–80 Hz	0.2 J × 40 Hz (8 W)

- Antiretropulsion devices improve SFR with semirigid URS. **Settings, not only the laser, decide outcome**.
- **Thermal** damage begins above **43 °C**, and **heat tracks power, not laser type**.
- Cap at **≤10 W in ureter** and ≤25 W in kidney; use a sheath and continuous irrigation, because stopping it causes thermal stricture.

6 · After URS

Most need neither long antibiotics nor a stent.

- Give a **single perioperative intravenous antibiotic dose**; extend only after urosepsis this episode.
- Stents cause symptoms in ~4/5; stentless uncomplicated URS is safe, yet >half are stented. String removal is preferred.

- Stent for perforation, stricture, tight ureter, infection or a second stage. After a sheath, inspect the ureter on withdrawal.

7 · SWL for ureteral calculi

Non-invasive, but limited by stone factors.

- SWL needs no general anaesthesia or stent and is an option at all levels.

Absolute contraindications	Relative
Pregnancy; uncontrolled bleeding disorder; anticoagulation that cannot stop (for treatment and 48 h); distal obstruction; untreated UTI; uncontrolled hypertension	Skeletal deformity or obesity preventing focusing; adjacent aneurysm

- **Predictors**: size matters most, and URS is better ≥10 mm. Impaction (CT "ring" sign, static stone) is possibly the strongest.
- Cystine, brushite and calcium oxalate monohydrate resist SWL. Density >1000 HU and skin-to-stone distance >10 cm are proven for renal stones only.
- **No routine pre-stenting**. Hold aspirin, anti-inflammatories (NSAIDs) and anticoagulants for treatment and 48 h. Give antibiotics only with stents or tubes, struvite or recent UTI.
- Technique *(Ch 98)*: upper ureter supine, mid-ureter over the sacrum **prone**, distal supine; **60–90 shocks/min** beats 120; ramp the energy.
- α-Blockers raise SFR.

8 · Special situations

Altered anatomy or pregnancy changes the route.

- **Pregnancy**: treat conservatively first; **SWL is contraindicated**; URS is safe with shielded fluoroscopy. Stents encrust, so change them **every 4 weeks**; use a retrograde stent before a nephrostomy.
- **Diversion or reimplant**: use SWL (prone) or **antegrade flexible URS**. Transplant: both routes are safe.
- Anomalies: escalate from SWL to PCNL to laparoscopic ureterolithotomy.

9 · Complications

Rates and immediate responses.

- **Perforation** (0–4%): stop, stent for ~4 weeks and drain any urinoma.
- An extruded stone is left alone and the patient told. A submucosal stone is lasered free and stented.
- Further lithotripsy embeds fragments, causing granuloma and stricture.
- **Avulsion** (0.06–0.5%) follows forceful extraction, mostly in the **proximal third**; place a **nephrostomy and repair later**, avoiding nephrectomy.
- **Stricture** occurs in 3–6% and is silent, so image every patient.
- **Sepsis** (~5%) is linked to positive culture, pre-stenting, diabetes and long surgery; treat the culture and keep IRP low.

KEY POINT

Nearly every serious URS complication is force without vision: blind basketing, forced sheaths, persistence in a tight ureter. Remedy: see, fragment, and when in doubt stop, stent and stage.

- Future: robotic URS and **burst wave lithotripsy** (handheld, in clinic).

10 · Landmark trials

Trial	Design	Result	Take-home
Stentless URS	RCT, stent vs none after uncomplicated URS	Safe without stent	Stent only for named indications
TFL vs Ho:YAG	RCT, low TFL settings	TFL faster; equal ureteral SFR	Low settings optimal for TFL

98 Surgical Management of Upper Urinary Tract Calculi

1 · Choosing the approach

The choice balances stone clearance against operative risk; this section sets the rules.

- The framework is the 2016 American Urological Association (AUA)/Endourological Society guideline, built on index patients and shared decision making. Pathogenesis *(Ch 95)*; ureteral stones *(Ch 97)*.

Domain	Factors
Stone	Size, number, location, composition
Renal anatomy	Obstruction or stasis, hydronephrosis, ureteropelvic junction obstruction (UPJO), calyceal diverticulum, horseshoe or ectopic kidney, lower pole
Patient	Infection, obesity, habitus deformity, coagulopathy, age, hypertension, renal failure or transplant, solitary kidney, diversion, pregnancy

KEY POINT

Only the stone domain is changed by choosing a modality; anatomy and patient factors decide which modality is available. Obesity, coagulopathy and pregnancy each exclude *a* modality, not surgery.

Asymptomatic renal stones

- Truly asymptomatic means no symptoms and no sequelae, including no recurrent urinary tract infection (UTI).
- The natural history is not benign: **~50%** have a stone event within **5 years**.
- **Size predicts intervention, not symptoms**; risk rises above 5 mm and above 10 mm.
- With **recurrent UTI and a non-obstructing stone**, remove it whatever its composition, and **culture the stone**.
- Treat residual fragments **> 4 mm** after ureteroscopy (URS).
- **Treat small asymptomatic contralateral stones** during surgery on the symptomatic side (see trials).
- If treatment is declined, give medical prevention *(Ch 99)* and serial imaging.

Preoperative evaluation

- AUA: non-contrast computed tomography (NCCT) **should** precede percutaneous nephrolithotomy (PCNL), and **may** guide the choice of shock wave lithotripsy (SWL) versus URS.
- SWL works best when density is **< 900 HU** (Hounsfield units) and skin-to-stone distance (SSD) is **< 10 cm**.
- Everyone needs urinalysis and culture within 1 month.

Modality selection

Scenario (AUA)	Offer	Caveat
<2 cm, non-lower pole, symptomatic	SWL or URS	SWL has fewer complications; URS has higher stone-free rate (SFR)
Total burden >2 cm	**PCNL**; staged URS an option	SWL not first-line

Scenario (AUA)	Offer	Caveat
Lower pole >1 cm	PCNL (highest SFR) or URS	SWL not first-line, because fragments clear poorly
Lower pole <1 cm	SWL or URS	—
Staghorn	PCNL	—
Cannot stop antiplatelet/anticoagulant	**URS**	Avoid SWL and PCNL

- European Association of Urology (EAU): for >20 mm use PCNL, then URS or SWL.
- EAU: for **10–20 mm, SWL and endourology are in equipoise**; for <10 mm use SWL or URS, then PCNL. Lower pole 10–20 mm is handled separately.

Stone morphometry scores

- P1 Stone complexity scores — *Guy's, STONE, CROES, S-ReSC predict SFR, not complications; Triple D predicts SWL*

2 · Shock wave lithotripsy

SWL is the least invasive option, so selection and technique decide success.

- P2 ESWL: mechanism, indications, factors affecting outcome, complications and current status — *peak **30–100 MPa**; sources; mechanisms (dynamic squeezing strongest); **60–90 shocks/min**, ramping, coupling; haematoma, steinstrasse (no routine stent); refuted effects; burst wave*

Not in the paper answer:

- Electromagnetic sources stay consistent up to 10^6 shocks; the electrohydraulic focus drifts up to 1 cm.
- Inability to target the stone is an absolute contraindication.
- The first human SWL was in 1980 (Dornier); the electrohydraulic HM3 remains the efficacy benchmark MCQ.
- **Absolute contraindications** are pregnancy, uncorrected coagulopathy or antithrombotics, untreated UTI, severe untreated hypertension, aortic or renal artery aneurysm (aorta > **4 cm**), and obstruction distal to the stone.
- No prophylactic antibiotic is needed if urinalysis is negative and there are no risk factors (AUA, EAU).
- For **arrhythmia**, gate shocks to the cardiac refractory period.
- †An obstructed or infected steinstrasse needs a nephrostomy or URS.

3 · Ureteroscopy (retrograde intrarenal surgery, RIRS)

URS is the workhorse for small renal stones and for patients on antithrombotics.

- Landmarks MCQ: Bozzini's Lichtleiter 1806 was the first endoscope; Takayasu placed the first ureteral access sheath (UAS) in 1974.
- Test the urine before elective URS, pre-treat bacteriuria, and give a single antibiotic dose chosen by the local antibiogram.
- **Antithrombotics**: URS is the modality for those who must continue them. Anticoagulation appears safe; antiplatelets increase admissions.

Instruments and principles

- **Safety wire**: AUA and EAU recommend it (EAU calls it controversial); a multicentre study found no outcome difference. Never force a wire past an impacted stone.
- The **UAS** lowers intrarenal pressure and eases repeated passes, but gives **no clear fall in infection**.
- The UAS injures the ureter in nearly half (severe in ~1 in 8), less if **pre-stented**; new stricture is rare.
- **Anti-retropulsion** devices (Stone Cone, NTrap, BackStop) reduce stone migration.
- AUA allows **stent omission** only when all four apply: normal anatomy, normal renal function, no ureteral injury and no planned second stage.

Success

The idea of "clinically insignificant fragments" is flawed, because small fragments cause reintervention in up to a fifth. Report zero-fragment rates.

Complications

Complication	Rate	Risk factors / management
Sepsis	~5%	Pre-stent, positive culture, ischaemic heart disease, age, long surgery, diabetes; treat the urine first
Subcapsular/perirenal haematoma	<1%	Hydronephrosis, thin cortex, long surgery, hypertension, UTI
Stricture	~3%	UAS injury, impacted stone; ultrasound surveillance (rate underestimated)
Perforation	Few %	Stop and stent
Avulsion	<1%	Proximal ureter; immediate nephrostomy, delayed reconstruction

4 · Percutaneous nephrolithotomy

PCNL clears the most stone in one session; safe access is the key skill.

- Landmarks MCQ: Fernström and Johansson 1976, first percutaneous stone removal; Wickham 1984, first tubeless; Valdivia-Uría 1987, first supine; Jackman 1998, first mini-perc (13 Fr).
- For 1–2 cm stones PCNL gives higher SFR and fewer procedures than SWL or URS. It is the treatment of choice for **staghorn** and most stones >2 cm.

PCNL first-line	Applications
Burden	Stones >2 cm, staghorn, matrix stones
Lower pole	>1 cm; lower pole diverticulum or infundibular stenosis
Poor retrograde route	Urinary diversion, inaccessible lower tract, prior ureteral stricture, injury or reconstruction
Coexisting pathology	UPJO; infected stones in non-ambulatory patients

Preoperative

- Stop anticoagulants with the prescriber (bridge if high risk); **low-dose aspirin is safe**.
- **NCCT is the gold standard**: it shows anatomy, infundibular length and angle, SSD, HU, and the ribs, pleura, spleen, liver and colon.

- For a poorly functioning kidney, get a renogram; the cut-off is debated at 10–20%, and nephrectomy may suit.
- Give single-dose, culture-directed prophylaxis.
- **Tranexamic acid** (Cochrane) reduces transfusion with no rise in thromboembolism; individualise it.

Positioning

- P1 P2 Prone vs Supine PCNL / patient positions in PCNL — *prone: upper pole access, ocular/brachial plexus risk; supine: airway, gravity drainage, ECIRS*

Access: the most important step

- Access obtained by the urologist gives shorter stay, fewer transfusions and fewer secondary procedures.
- The **ideal** target is a posterior calyx with a short, wide infundibulum.
- An upper pole posterior calyx reaches most of the system, but **supracostal (above the 12th rib)** access raises chest complications.
- Lower pole access risks more colonic injury (left side), and a steep infundibulopelvic angle hampers reaching the UPJ.

Step	Eye-of-the-needle (bulls-eye)	Triangulation
Align	C-arm rotated 15–30° toward surgeon; needle seen end-on	Mediolateral in the oblique view, cephalocaudal in the anteroposterior (AP) view
Depth	Rotate back past AP	Short advances, alternating views

- **Air** fills non-dependent calyces but blocks ultrasound. Confirm entry by urine return, not contrast.
- **Ultrasound access** uses no radiation and shows bowel, spleen, liver and pleura; stones show by shadow and twinkle artefact.
- In **endoscopic combined intrarenal surgery (ECIRS)**, a ureteroscope watches the puncture, giving fewer tracts, less bleeding and higher clearance.

Dilatation and tract size

- Dilate with Alken metal dilators (dense scar), Amplatz serial dilators, or a **balloon** (one pass, 20–30 atm). Keep a safety wire unless there is a through-and-through wire.
- Multiple tracts raise bleeding and reduce function, especially in chronic kidney disease.
- P4 Miniperc: miniaturised PCNL, its nomenclature and newer PCNL techniques — *standard 24–34, mini 15–22, ultra-mini 11–14, micro <10 Fr; mini: equal SFR, less bleeding*

Completion and exit

- Finish with **flexible nephroscopy or URS**; a randomised trial showed higher SFR, fewer tracts and less bleeding.

ALGORITHM

1. Confirm the collecting system and ureter are clear.
2. Withdraw the sheath under vision, with the safety wire out last.
3. Compress the tract.
4. If bleeding persists, re-inflate the balloon or place a **Kaye tamponade catheter**, and alert the anaesthetist.
5. If it still persists, arrange **selective angioembolisation**.

- Drain with a nephrostomy (≤10 Fr means less pain), a double-J stent, or go tubeless. Fibrin glue does not reduce bleeding.

Complications

- Cochrane (42 trials): **PCNL equals RIRS for major complications**, with fewer secondary procedures.
- Complications include haemorrhage (†delayed bleeding from arteriovenous fistula or pseudoaneurysm needs angioembolisation), visceral injury, and hydrothorax or pneumothorax (†chest drain if significant).
- Sepsis, gas embolism and death also occur.

5 · Special anatomy

Abnormal kidneys drain poorly and are reached differently.

Anatomy	Key adaptation
Horseshoe (stones in ~1 in 5)	SWL: long SSD. PCNL: access is medial and caudal; the upper pole posterior calyx is usually subcostal; more colonic injury; UPJO is frequent
Calyceal diverticulum/infundibular stenosis	Treat only if pain, UTI or haematuria. URS for upper or interpolar (laser-incise or dilate the ostium); lower pole: PCNL. Fulguration controversial
Pelvic kidney	SWL or URS <2 cm; PCNL with ultrasound, with or without CT or laparoscopic guidance
Urinary diversion	Antegrade wire access most reliable
Transplant	Always remove stones. The denervated graft presents with fever, oliguria and haematuria, not colic. Ultrasound-guided PCNL
Spinal deformity	SWL poor; URS good; PCNL equal clearance but more infection

- **Laparoscopic or robotic** surgery is only for when SWL, URS and PCNL fail or are expected to. With UPJO, clear the stones and then reassess the junction.

6 · Landmark trials

Trial	Design	Result	Take-home
Lower Pole I	Randomised controlled trial (RCT), PCNL vs SWL, lower pole >1 cm	SFR 91% vs 21%	SWL not first-line for lower pole >1 cm
Contralateral stone RCT (2022)	Remove vs observe small asymptomatic stones during surgery	Relapse 16% vs 63%	Treat contralateral stones
Cochrane PCNL vs RIRS (2023)	42 trials	No difference in major complications; fewer secondary procedures	PCNL no longer "more morbid"

99 Prevention of Urinary Stone Disease

1 · Stone disease as a chronic condition

Recurrence is the disease, so prevention is lifelong.

- Urinary stone disease is a **chronic metabolic disorder punctuated by colic**. Pathogenesis *(Ch 95)*.
- Up to **50%** recur within 5–10 years. The **ROKS** nomogram predicts a second stone, and ROKS 2 predicts multiple recurrences.
- **Low urine volume** is the single most important risk; others are young age, male sex, obesity, family history and low citrate.

2 · Screening and metabolic evaluation

Cheap screening sorts every first-time former into low or high risk.

- P1 P2 Metabolic Evaluation in Stone Disease — *screening (history, blood, urine, stone analysis, imaging) for every first-time former; high risk: one or two 24-hour urines; fast/calcium-load testing discouraged*

Not in the paper answer:

- Some prefer two 24-hour collections, because paired samples often differ.
- A 24-hour urine is **not** part of screening.
- Lithogenic drugs: calcium, vitamins C and D, acetazolamide, steroids.

3 · General measures (all formers)

Cheap measures for every stone type.

- **Fluid** is the cornerstone: aim for urine ≥**2.5 L/day** (about 3 L intake).
- **Limit nondairy animal protein**, because it raises urinary calcium and uric acid and lowers citrate.
- **Keep sodium low**, because sodium raises calcium excretion and lowers citrate. The Dietary Approaches to Stop Hypertension (DASH) diet lowers risk.
- Keep **calcium intake normal**, because restriction *increases* stones. Eat more fruit and vegetables.

> **KEY POINT**
> Why normal calcium? Dietary calcium binds oxalate in the gut; restricting it frees oxalate for absorption.

4 · Hypercalciuria: thiazides

Thiazides are the main drug for calcium stone recurrence.

- The American Urological Association (AUA) makes thiazides the **preferred drug** for hypercalciuria with recurrent calcium stones, **and** for recurrent calcium stones with a normal evaluation. Yendt 1966 MCQ.
- **Hypokalaemia causes intracellular acidosis, which leads to hypocitraturia**. So add **potassium citrate**, and check potassium 1–2 weeks after starting or changing the dose.
- Amiloride with a thiazide adds calcium-lowering effect, so **never add potassium** to it. Avoid triamterene, as it forms stones.
- Thiazides raise bone density; side effects include gout, diabetes and sexual dysfunction.

Target	Drug and dose	Key side effects
Calcium	Hydrochlorothiazide 25 mg twice daily; chlorthalidone 25–50 mg; indapamide 2.5 mg	Hypokalaemia, hyperuricosuria, hypocitraturia
Citrate	Potassium citrate 30–60 mEq/day	Gastrointestinal (GI) upset, hyperkalaemia
Uric acid	Allopurinol 300 mg	Rash, myalgia, hypersensitivity
Oxalate	Magnesium; pyridoxine 100 mg	Diarrhoea (magnesium); ataxia, numbness (pyridoxine)
Cystine	Tiopronin; D-penicillamine 250 mg titrated	See thiol table
Urease	Acetohydroxamic acid 250 mg three times daily	Heaviest burden (below)

5 · Hyperoxaluria

Find the source of oxalate: diet, gut or liver.

- Stone risk rises continuously with urinary oxalate. Avoid large portions of spinach, nuts, chocolate and tea.
- **Enteric** hyperoxaluria follows malabsorption, diarrhoea or Roux-en-Y gastric bypass; urine oxalate is >40, often 60–70+ mg/day.
- In it, unabsorbed fat binds (saponifies) calcium, which frees oxalate for absorption in the colon.
- Treat with **calcium up to 1 g/day with meals**, very high fluid, **liquid** potassium citrate (wax tablets pass unabsorbed), magnesium and a low-oxalate diet.
- P2 Primary Hyperoxaluria — *oxalate often >**100 mg/day**; PH1 (~70%, AGXT, pyridoxine-responsive), PH2 (GRHPR), PH3 (HOGA1); lumasiran silences glycolate oxidase (ILLUMINATE-A)*

Not in the paper answer:

- Primary hyperoxaluria type 3 (PH3) is the mildest and often has hypercalciuria.
- Lumasiran is approved for PH1.

6 · Hypocitraturia

Citrate is the main urinary inhibitor.

- AUA offers **potassium citrate** 30–60 mEq/day for recurrent calcium stones with low citrate, **and** for those with no abnormality.
- Sodium citrate does not lower urinary calcium. Use caution in renal impairment and with potassium-sparing diuretics.
- Over-alkalinising to **pH ≥7** creates **calcium phosphate** risk.
- **Alkali citrate, not total citrate, matters**. Orange juice also raises oxalate, and the Moonstone supplement is acidic MCQ.

7 · Uric acid stones: a pH disease

Acid urine is the usual cause, so alkali comes first.

- P2 Uric acid calculi: etiopathogenesis, medical management and allopurinol — *pKa **5.5**; low pH the main cause; potassium citrate to **6.0–6.5**, avoid >7.0; allopurinol not first-line; hypersensitivity (fatal up to 27%)*
- Sodium alkali risks calcium stones.

- Give allopurinol only for hyperuricaemia with hyperuricosuria, or failed alkalinisation; its risk is higher with thiazides.

8 · Cystine stones

Treatment climbs a ladder.

ALGORITHM

1. **Dilute**: urine ≥ **3 L/day**, including at night; sodium ≤2300 mg/day; moderate protein (not in children).
2. Aim for cystine < **250 mg/L**.
3. If not enough, **alkalinise** with potassium citrate to pH **7.0–7.5**.
4. If still forming stones, add a **thiol** (tiopronin).

- Cystine's pKa of 8.3 cannot be reached, and pH >7.5 risks calcium phosphate.
- Sodium alkali raises cystine excretion.
- Thiols form a soluble cystine–drug disulfide.

	D-Penicillamine	Tiopronin	Captopril
Status	First thiol; largely replaced	**Thiol of choice**; fewer stop it	Weak evidence
Dose	250 mg/day, titrate	300 mg twice daily, up to 1200 mg/day	—
Toxicity	Rash, fever, arthralgia, pancytopenia, nephrotic syndrome	Asthenia, GI upset, rash, joint aches	—

9 · Infection (struvite) stones

Struvite is a surgical disease first.

- **Clear the stone completely** (percutaneous nephrolithotomy for staghorn *(Ch 98)*), because every fragment harbours bacteria.
- **Acetohydroxamic acid (AHA)** is a urease inhibitor. AUA: use it only for residual or recurrent struvite after surgical options are exhausted.
- AHA toxicity is ~30%: haemolytic anaemia, **deep venous thrombosis** (most severe), tremor and headache.
- AHA is contraindicated if creatinine is > **2.5 mg/dL** or in pregnancy.
- Most struvite formers are also hypercalciuric or hypocitraturic, so do a 24-hour urine.

10 · Monitoring

Check that treatment works and is tolerated.

ALGORITHM

1. Start therapy.
2. Repeat the 24-hour urine within **6 months**.
3. Then check urine and blood (for side effects) at least yearly.
4. If a new stone forms, **re-analyse** it.
5. Image periodically.

11 · Landmark trials

Trial	Design	Result	Take-home
Borghi 1996	Randomised controlled trial (RCT), high fluid, calcium oxalate formers	5-year recurrence 12% vs 27%	Urine >2 L/day prevents stones
Borghi 2002	RCT, hypercalciuric formers	Low protein and salt with normal calcium halved events vs low calcium	Never restrict calcium
Ettinger 1988	Double-blind RCT, chlorthalidone	~90% fewer stones	Thiazides prevent calcium stones
Barcelo 1993	RCT, potassium citrate	Fewer new stones at 3 years	Citrate prevents recurrence
ILLUMINATE-A	Phase 3 RCT, lumasiran, PH1	84% vs 0% reached near-normal urine oxalate	RNA interference for PH1

100 Lower Urinary Tract Calculi

1 · Bladder calculi

The stone type tells you whether to seek a bladder cause.

- Bladder stones are ~5% of urinary stones. Types are **migrant**, **primary** (endemic) and **secondary**.

	Primary (endemic)	Secondary
Patient	Children; peak **2–4 years**; boys far more than girls	Men >60 with outlet obstruction
Cause	Cereal diet low in animal protein and phosphate, causing low urinary phosphate	Bladder outlet obstruction (BOO), foreign body, catheter, infection
Composition	Ammonium acid urate, calcium oxalate, uric acid	**Struvite**
Recurrence	Solitary; rarely recurs	Recurs unless the cause is removed

- Foreign bodies act as a nidus (**heterogeneous nucleation**): a migrated intrauterine device, sutures, Foley balloon fragments, or hair during clean intermittent catheterisation (CIC).
- Drugs: triamterene, indinavir.

> **KEY POINT**
> Primary stone is a disease of the urine (normal bladder); secondary is a disease of the bladder (stasis, nidus, infection). Do not hunt for outlet obstruction in the endemic child.

- Adults present with **terminal haematuria** and lower urinary tract symptoms. Children have sandy urine and dysuria.

2 · Management of bladder stones

Choose by the safest route into the bladder.

The route, not the stone, dictates modality.

Option	Indication / key point
Chemodissolution	Adjunct only; for catheter encrustation
Extracorporeal shock wave lithotripsy (ESWL)	Artificial sphincter, penile prosthesis, neobladder, high-risk patient; works poorly in continent reservoirs
Transurethral cystolithotripsy	Most popular; holmium laser of choice; risks urethral stricture
Percutaneous cystolithotomy	Difficult urethra (bladder neck reconstruction or closure); spares the urethra
Open cystolithotomy	Small-capacity bladder, stricture, no safe endoscopic route

- Cystolithotomy means intact removal; cystolitholapaxy means mechanical crushing.

3 · Special situations

Treat the cause along with the stone.

- **BOO** is present in 45–79%. **A bladder stone secondary to BOO is an absolute indication for benign prostatic hyperplasia (BPH) surgery**.
- But >30% are infected and some have no BOO, so **prove BOO** first.

- **Diversion or augmentation**: hyperchloraemic acidosis causes hypercalciuria; there is enteric hyperoxaluria; urease producers form struvite; and **mucus** forms a biofilm.
- A conduit with stomal stenosis: clear the stones and revise the stoma. A **continent pouch: go percutaneous**, which protects continence. A neobladder: transurethral, or percutaneous or open if large.
- **Spinal cord injury**: risk peaks **3 months** after injury; 15–30% form stones within 10 years.
- **CIC beats indwelling catheters**; suprapubic equals urethral.
- Stones ≤4 cm get transurethral laser; larger ones percutaneous or open. Use ESWL for anaesthetic risk or autonomic dysreflexia.
- **Transplant**: **remove all exposed suture**.
- **Children**: the urethra is small, so use percutaneous cystolithotripsy for large stones, even in infants.

4 · Urethral calculi

Site sets the treatment.

- Urethral stones are 0.3–1% of stones, in males. Most are **migratory** (calcium oxalate, acute).
- A minority are primary struvite, from stricture, diverticulum or hair-bearing graft.
- The commonest site is the posterior urethra (up to 88%).
- They cause **acute painful retention**. Posterior stones give perineal or rectal pain; anterior stones give penile-tip pain and a palpable mass. ~98% are radio-opaque.

Site	Treatment
Posterior	**Push back** into the bladder, then laser; else treat in situ
Anterior	**No pushback**; in situ holmium laser; urethrotomy for large stones
Fossa navicularis	Forceps; ventral meatotomy
Diverticular / stricture	Extraction with diverticulectomy or stricture repair, same sitting or staged

5 · Unusual sites

Rare stones, one key point each.

- **Preputial** stones follow phimosis and poor hygiene and are struvite. Treat by circumcision, and send the **excised skin to histopathology**.
- **Prostatic** stones are calcium phosphate, forming in acini (endogenous) or from urine reflux (exogenous). They are linked to BPH, **not cancer**. Transrectal ultrasound (TRUS) is the main tool.
- **Seminal vesicle** stones cause haematospermia and painful ejaculation; **TRUS is most accurate**.
- Treat by vesiculectomy or transurethral resection of the ejaculatory ducts; seminal vesiculoscopy has fewer complications.

54 Management of Paediatric Kidney Stone Disease

1 · Epidemiology

Children carry stone disease for life, so kidney, bone and radiation risks add up.

- Early onset means stones before **24 years**.
- Incidence is rising fastest in **adolescent girls**, and ~50% recur within 3–5 years.

Feature	Children	Adults
Sex ratio	Female predominance	Male (historically 3 : 1)
Calcium oxalate (CaOx)	≈ **80%**, mostly idiopathic, on Randall's plaques	75–80%
Calcium phosphate (CaP) / uric acid	CaP commoner and uric acid rarer, because urine is more alkaline	≈ 5% each
Lifetime stakes	Chronic kidney disease (CKD), hypertension, low bone mineral density (BMD); cumulative radiation	Shorter

- **High-risk** groups are bladder augmentation (≈ 1%/year), malabsorption, tube feeds and immobility.

> **KEY POINT**
> Stone formers get more CKD, hypertension and fracture. Childhood is the bone-accrual window, so seek low BMD early.

2 · Diagnosis and imaging

Ultrasound first, to keep radiation low.

> **ALGORITHM**
> 1. If a stone is suspected, do grey-scale **ultrasound (US)** with colour Doppler.
> 2. If US is negative but suspicion is high, or there is hydronephrosis, then do **low-dose non-contrast computed tomography (NCCT)**.
> 3. If US is negative and suspicion is low, stop.

- US shows the **twinkle artefact**; sensitivity is > 70% and specificity > 95%, but it overestimates size by 1–2 mm.
- NCCT is ≈ 100% accurate; use 40 mA for children ≤ 50 kg.

3 · Metabolic evaluation: every child

Most children have a metabolic cause, so all are evaluated.

An abnormality raises recurrence up to **fivefold**, so do a **full evaluation after the first stone**.

- **History**: ketogenic diet, vitamins C and D, furosemide, acetazolamide, topiramate, prematurity.
- **Examination clues**: dysmorphism suggests Williams syndrome.
- Rickets suggests Dent disease or hereditary hypophosphataemic rickets with hypercalciuria (HHRH).
- Tetany suggests familial hypomagnesaemia with hypercalciuria and nephrocalcinosis (FHHNC).

- Do **stone analysis** first: cystine (hexagonal crystals), struvite, 2,8-dihydroxyadenine and xanthine are each diagnostic.
- Measure parathyroid hormone only with hypercalcaemia and hypophosphataemia.
- With hypercalcaemia and hypercalciuria, check the vitamin D metabolite ratio for **CYP24A1** loss, which mimics vitamin D intoxication.
- In proteinuric boys, check urine β2-microglobulin for Dent disease.
- If not toilet-trained, use spot solute : creatinine ratios.

Metabolite	Normal (child)
Calcium	< **4 mg/kg/day**; spot Ca : creatinine < 0.8 (0–6 mo), < 0.6 (7–12 mo), < **0.21** (> 2 y)
Oxalate	< 40–50 mg/1.73 m²/day
Citrate (lower limit)	Hypocitraturia is < 128 mg/g creatinine (male), < 300 (female)
Cystine	< 50–60 mg/1.73 m²/day
Uric acid	< 815 mg/1.73 m²/day; spot < 0.56 mg/dL glomerular filtration rate

KEY POINT

Never use adult limits: infants excrete more calcium, so the school-age 0.21 ratio overcalls them. Citrate's abnormality is too *little*.

4 · Urinary abnormalities

Name the abnormality, because each has its own treatment and genetic causes.

Hypercalciuria: commonest (30–50%)

Serum calcium	Causes
High	Hyperparathyroidism, CYP24A1, hypervitaminosis D, immobilisation, Williams
Low	Hypoparathyroidism; autosomal dominant hypocalcaemic hypercalciuria (calcium-sensing receptor gain)
Normal	Prematurity, loop diuretics, topiramate, ketogenic diet, Dent, Bartter, FHHNC, distal renal tubular acidosis (dRTA), HHRH; if none, **idiopathic**

Hyperoxaluria

- P2 Primary Hyperoxaluria — *urinary glycolate/L-glycerate, then genetics; pyridoxine*

Not in the paper answer:

- Only 10–30% of primary hyperoxaluria type 1 (PH1) respond to pyridoxine.
- Thiols and lumasiran are the only drugs approved for paediatric stone prevention.
- **Secondary** hyperoxaluria follows fat malabsorption, a low-calcium diet or vitamin C.

Others

- **Hypocitraturia**: incomplete dRTA needs an acid-load test.
- **Cystinuria** (*SLC3A1*/*SLC7A9*) may present in infancy with a staghorn.
- **Hyperuricosuria** causes: hypoxanthine–guanine phosphoribosyltransferase (HPRT) deficiency (Lesch–Nyhan), phosphoribosyl pyrophosphate synthetase (PRPS) superactivity, *SLC22A12*.

Monogenic disease	Inheritance	Signature
Dent (*CLCN5*)	X-linked	Low-molecular-weight proteinuria, nephrocalcinosis, CKD

Monogenic disease	Inheritance	Signature
Bartter	Recessive	Salt wasting, hypokalaemic **alkalosis**, hypercalciuria
FHHNC (*CLDN16/19*)	Recessive	Hypomagnesaemia, tetany, nephrocalcinosis
dRTA	Dominant/recessive	Hypokalaemic **acidosis**, hypocitraturia, nephrocalcinosis

5 · Management principles

Guidelines mirror adult practice, with smaller instruments and less radiation.

- Tamsulosin is off-label; use it for ≤ 6 weeks. Up to 60% need surgery.
- The residual fragment threshold is **4 mm on US**: 2 mm on CT plus the ≈ 2 mm US overestimate.
- Give antibiotic prophylaxis for all ureteroscopy (URS) and percutaneous nephrolithotomy (PCNL), but for shockwave lithotripsy (SWL) only with infection risk; limit it to ≤ 24 h.

Scenario (American Urological Association / Endourological Society 2016)	Recommendation
Uncomplicated ureteric ≤ **10 mm**	Observation with or without an α-blocker
Ureteric, failed or unlikely to pass	URS or SWL (strong)
Before URS	No routine pre-stenting
Renal ≤ **20 mm**	SWL or URS
Renal > 20 mm	PCNL or SWL (with stent or nephrostomy); authors prefer PCNL
Before PCNL	Low-dose CT (strong)
Asymptomatic non-obstructing renal	Surveillance with US
Open/laparoscopic/robotic	Only with anatomic abnormality

6 · Ureteroscopy

The small ureter is the main limit, so never force access.

- Clearance is ≈ 90%, but worse under 6 years, under 20 kg and in the lower pole.
- Scope sizes (French, Fr): semirigid 4.5–6, flexible 7.5–9.

ALGORITHM

1. Confirm a sterile culture; place in lithotomy; do cystoscopy.
2. Do a retrograde pyelogram and place a safety wire.
3. Calibrate the orifice with an 8/10 Fr coaxial dilator.
4. **If it will not pass, then place a 4.6–4.8 Fr stent for ≥ 14 days and return later.**
5. If it passes, do URS.
6. Stent if the case was long, needed many passes, or caused trauma or oedema.

- **No balloon or active orifice dilatation**; rely on passive stent dilatation.

7 · Shockwave lithotripsy

Children suit SWL because the skin-to-stone distance is short.

- SWL equals URS for small stones.

- Favourable: < 10 mm, non-lower-pole, < 1000 Hounsfield units. Unfavourable: cystine, brushite, infundibulopelvic angle > 45°.
- Use < 20 kV and ≤ 2000–3000 shocks (< 2000 in infants); **80/min is better than 120/min**.
- Absolute contraindications are bleeding diathesis, untreated urinary infection (UTI) and pregnancy.

8 · Percutaneous nephrolithotomy

PCNL is first-line for large stones, but tract size must shrink for a child.

- PCNL is first-line for stones > 20 mm.
- A 24–30 Fr tract in a child equals ≈ 72 Fr in an adult, so **miniaturise** (mini < 20–24, micro < 10 Fr).

ALGORITHM

1. Prevent hypothermia; place a ureteric catheter.
2. Position prone (latex precautions in myelomeningocele).
3. Puncture a posterior calyx under US or fluoroscopy.
4. Dilate serially to an 18 Fr sheath (24 Fr for staghorn).
5. Fragment with nephroscope and ultrasonic or laser energy; check with a flexible scope.
6. Place a nephrostomy (7 Fr infant to 16 Fr adolescent).

- Transfusion is needed in < 10%; late bleeding is treated by angioembolisation.

9 · Secondary prevention

Lifelong prevention matters more in children than in adults.

- **Fluid**: at least maintenance; > 2–2.5 L/day in adolescents; 1.5–2 L/m² in cystinuria or PH.
- **Sodium** < 2–3 mEq/kg/day. **Calcium is not restricted**, because it binds gut oxalate.
- Target urine pH > 6.5 for uric acid and > 7 for cystine; avoid > 6.5 in calcium formers.

Drug	Indication	Child dose / caution
Hydrochlorothiazide	Hypercalciuria despite salt restriction	1–2 mg/kg/day; for hypokalaemia add potassium citrate
Potassium citrate	Hypocitraturia, uric acid, cystine	2–4 mEq/kg/day; CaP risk if pH > 6.5
Tiopronin	Cystinuria failing fluid and alkali	Fewer adverse effects than penicillamine
Allopurinol	HPRT, PRPS, adenine phosphoribosyltransferase deficiency; hyperuricosuric CaOx	4–10 mg/kg/day; can cause xanthine stones
Pyridoxine	PH1 (trial in all)	10–30% respond
Lumasiran	PH1	RNA interference against glycolate oxidase

† PH1 with end-stage renal disease needs combined liver–kidney transplantation, because kidney-alone grafts fail.

10 · Landmark trials

Trial	Design	Result	Take-home
ILLUMINATE-A	Randomised, lumasiran vs placebo, PH1	Urinary oxalate lowered by 65%	RNA interference therapy for PH1

125 Genital and Lower Urinary Tract Trauma

1 · Principles

Why this matters: the first decision decides long-term function.

- Lower tract injuries rarely threaten life; disability comes from mismanagement, not the injury.
- Key decision: **immediate reconstruction vs deliberate delay**. Testicular rupture is repaired at once; pelvic fracture urethral injury (PFUI) gets a suprapubic tube now and urethroplasty later.
- Imaging: ultrasonography for scrotum and equivocal penile fracture; computed tomography (CT) cystography for bladder; retrograde urethrography (RUG) for urethra.

2 · Penile fracture

Why this matters: diagnosis is clinical, repair is urgent, and the urethra must be checked.

- P2 Fracture penis — *tunica thinnest ventrolaterally; snap, then detumescence, then eggplant deformity; deviates away from tear; urethral injury ~20% West vs ~3% Middle East; clinical diagnosis, ultrasound then magnetic resonance imaging (MRI); intraoperative flexible cystoscopy; delay ≤7 days harmless; complications ~20%*

Not in the paper answer:

- Bilateral tears (~10%) carry more urethral injury, and absence of blood at the meatus does not exclude it.
- Collagenase ruptures mostly occur after the 14-day observation window; manage them conservatively unless the urethra is injured.
- Avoid deep corporal ligation as well as excessive debridement of erectile tissue.

ALGORITHM

1. Explore promptly. Through a ventral penoscrotal (or small lateral) incision, evacuate haematoma and find the tear (artificial erection).
2. Irrigate; close the tunica with interrupted **2-0/3-0 absorbable** sutures.
3. If the urethra is injured, repair it.

- A breached Buck's fascia gives a †butterfly haematoma. A negative MRI reliably excludes a fracture. With a distal circumcising incision, add circumcision if uncircumcised.

3 · Penetrating, bite, amputation, zipper and strangulation

Why this matters: each mechanism has its own rule.

- **Gunshot**: explore immediately, not delaying for imaging. Irrigate, remove foreign matter, give antibiotics and close the tunica, confirming closure with an artificial erection. These wounds are seldom isolated.
- Urethral injury occurs in **15–50%** of penile gunshots, so check with RUG or intraoperative dye. Viable tissue is repaired primarily; high-velocity tissue loss needs suprapubic diversion and staged repair.
- **Dog bites**: irrigate, debride and close **primarily**, with amoxicillin–clavulanate and tetanus and rabies cover. **Human bites** present late and infected, so **leave them open** and give amoxicillin–clavulanate or moxifloxacin.

Penile amputation

- It is usually self-mutilation, so **always get psychiatric consultation**. **Microsurgical replantation** is preferred; otherwise repair urethra and corpora macroscopically.
- Transport the part **"double bagged"**: saline gauze in a sterile bag, inside an outer bag of ice slush. Long ischaemia time alone is no reason to abandon replantation.
- Erections return in >50% with either technique; microvascular repair gives less stricture, skin and sensory loss. Part unavailable: formalise the stump. Total loss: radial forearm phalloplasty.

ALGORITHM

1. Suprapubic cystostomy; debride.
2. Tunica with 2-0 absorbable; urethra over a catheter with 4-0.
3. Anastomose dorsal artery with **11-0**, dorsal vein with **9-0** and dorsal nerve with **10-0** nylon.
4. Provide skin cover.

- **Zipper** injury (children): use a penile block and mineral oil and make one attempt. If that fails, cut the cloth between the teeth or the slider's **median bar** with bone cutters; circumcise if needed.
- **Strangulation**: in children think of hair tourniquet and **child abuse**. Escalate from lubrication and removal, to the **string technique** (tape under the ring, wound distally, unwound from the proximal end), to glanular puncture, to cutting tools with saline cooling in theatre. Suprapubic catheter if retention.

4 · Testicular injury

Why this matters: early exploration saves the testis, and imaging must not delay it.

- Scrotal mobility, the cremasteric reflex and a tough tunica protect the testis. About 80% of penetrating scrotal wounds also involve neighbours (thigh, penis, urethra, **femoral vessels**).
- Consider rupture in every blunt injury; haematoma size does not predict it. **Ultrasonography** sensitivity is only ~64–75%, so an **equivocal scan must not delay exploration**.
- A non-palpable testis after trauma suggests **dislocation**, commonest to the superficial inguinal pouch (50%). Trauma does not exclude torsion.

†American Association for the Surgery of Trauma (AAST) testis grade	Injury
I	Contusion/haematoma
II	Subclinical laceration of tunica albuginea
III	Tunical laceration with <50% parenchymal loss
IV	Major laceration with ≥50% parenchymal loss
V	Total destruction or avulsion

ALGORITHM

1. Explore; remove necrotic and extruded tubules.
2. Close the tunica with small absorbable sutures.
3. If the defect is large, use a tunica vaginalis flap or graft (never synthetic); drain.

- Explore a **significant haematocele** (up to 80% hide rupture) and an **intratesticular haematoma** (delayed orchidectomy 15%).
- Always explore penetrating injury; ligate an injured vas (non-absorbable), reconstruct later, and consider contralateral exploration. Close even small defects, because rising pressure extrudes tubules.
- Salvage is **>90%** if repaired within **72 h**; non-operative care leads to infection, atrophy and late orchidectomy.

5 · Genital skin loss

Why this matters: penis and scrotum need different grafts.

- Causes: Fournier gangrene debridement (commonest), avulsion, burns, blast. Fournier gangrene shows crepitus and subcutaneous air.
- Treat Fournier gangrene with antibiotics, resuscitation and **repeated debridement**, then negative pressure wound therapy with daily inspection. Divert early with a suprapubic tube if extensive; hyperbaric oxygen is unproven.
- Deep electrical burns are managed conservatively.

	Penis	Scrotum
Graft	**Thick, non-meshed** split-thickness skin graft (STSG) 0.012–0.015 inch, because meshed graft contracts	**Meshed** STSG after local flaps (exudate escapes, so better take)
Key step	Excise residual subcoronal skin, which otherwise causes lymphoedema; never reuse avulsed skin	Suture the testes together to avoid a bifid scrotum; ≤50% loss closes primarily
Note	Graft is insensate, but sexual function is preserved	Thigh pouches not recommended

- Graft 5–7 days after the last debridement (or after ≥72 h culture-directed antibiotics, granulation and a normal white count), under a bolster or vacuum dressing.

6 · Bladder injury

Why this matters: intraperitoneal tears need surgery, while most extraperitoneal tears heal on a catheter.

- P2 Lower urinary tract trauma in pelvic fracture: classification, assessment and (early) management of bladder and urethral injuries — *pelvic fracture in 83–95% of bladder injuries, bladder in 5–10% of fractures; cystography for gross haematuria and pelvic fracture; retrograde fill **350 mL**, drainage film (plain only); flame-shaped extraperitoneal; 22-Fr catheter for uncomplicated extraperitoneal; intraperitoneal needs immediate repair; suprapubic tube adds nothing*

Not in the paper answer:

- Iatrogenic injury is commonest in obstetric and gynaecological surgery. Penetrating pelvic or buttock injury with any haematuria needs cystography.
- Underfilling the bladder gives false-negative cystograms.
- Poor catheter drainage is a catheter problem until proven otherwise. Bone spicules in the wall prevent healing, so repair them.
- A ureteric orifice injury needs a stented reimplant.

†AAST bladder grade	Injury
I	Contusion, intramural haematoma, partial-thickness laceration
II	Extraperitoneal laceration <2 cm

†AAST bladder grade	Injury
III	Extraperitoneal ≥2 cm, or intraperitoneal <2 cm
IV	Intraperitoneal laceration ≥2 cm
V	Laceration extending into bladder neck or ureteric orifice (trigone)

- Intraperitoneal tears are larger than imaging suggests. Leaked urine causes chemical peritonitis and, once reabsorbed, **acidosis and azotaemia**. Laparoscopic repair suits selected isolated cases.

KEY POINT

Immediate repair: intraperitoneal, penetrating, bladder neck, rectal/vaginal injury, bone fragments in bladder, clots/poor drainage, iatrogenic urological injury, laparotomy for other reasons.

7 · Posterior urethral injury (PFUI)

Why this matters: drain now and reconstruct later.

- P2 P3 Lower urinary tract trauma in pelvic fracture / early and late management of PFUD — *bulbomembranous junction; triad blood at meatus, retention, palpable bladder; RUG technique; suprapubic tube standard, brief realignment optional (no difference); delayed excision and primary anastomosis at **2–3 months** (corporal split, then pubectomy, then rerouting); erectile dysfunction (ED) up to 82%, restenosis 5–15%, incontinence <4%*

Not in the paper answer:

- The AAST urethral grades (table below) are not in the answer.
- ED is predicted by symphyseal diastasis, lateral displacement and a long gap. Consider penile duplex Doppler, because ischaemia may predispose to restenosis.
- PFUI complicates ~10% of male and up to 6% of female pelvic fractures, most with **straddle (four-rami)** and unstable fractures. The sphincter is usually preserved, so continence is often maintained. A "high-riding" prostate is unreliable.
- Females: vaginal examination in every pelvic fracture; urethroscopy preferred. The scarred gap is a **stenosis**, not a stricture.
- A suprapubic tube is safe with pelvic hardware if tunnelled high in the midline. A partial tear allows one gentle catheter attempt.
- After realignment, remove the catheter at 4–6 weeks but keep the suprapubic tube. Endoscopic "cut-to-the-light" fails for gaps ≥1 cm. Limit lithotomy to ≤5 h.

†AAST urethra grade	Injury
I	Contusion — blood at meatus, normal urethrogram
II	Stretch — elongation without extravasation
III	Partial disruption — extravasation, contrast reaches bladder
IV	Complete disruption, <2 cm separation
V	Complete, >2 cm separation or extension into prostate/vagina

ALGORITHM

1. If there is blood at the meatus, do RUG before any catheter.
2. Place a 16-Fr catheter in the fossa navicularis and inflate the balloon with 3 mL.
3. Apply penile traction, position oblique, inject 20 mL contrast and image during injection.

8 · Anterior urethral injury

Why this matters: a straddle injury can cause a stricture years later.

- Most are **straddle injuries** crushing the bulbar urethra against the pubis. Contusion or incomplete tear needs a catheter; major injury a suprapubic tube.
- Low-velocity gunshot wounds are repaired primarily; high-velocity wounds are diverted. The spongiosum has a good blood supply, so debride it minimally.
- Later, the 1.5–2 cm obliterated bulbar segment is treated by **excision and primary anastomosis** (>95% success). Urethrotomy through an obliteration is futile.

9 · Landmark trials

Trial	Design	Result	Take-home
Multi-institutional PFUI study	Prospective, suprapubic tube vs early endoscopic realignment	No difference in stenosis or urethroplasty need	Suprapubic tube remains standard; realignment optional, brief

62 Paediatric Genitourinary Trauma

1 · Renal injury: how the child differs

Why this matters: normal blood pressure and clear urine can hide major injury.

- The kidney is the commonest site of paediatric genitourinary (GU) trauma. Most injuries are **blunt** and grade 1–3; half have other injuries.
- Nephrectomy (3–5%) is **more likely in adult than paediatric trauma units**. Congenital anomalies (e.g. ureteropelvic junction [UPJ] obstruction) are 3–5 × commoner than in adults.

Feature	Child	Adult
Protection	Less perirenal fat, weaker wall, kidney below the ribs	Better cushioned
Blood pressure	Maintained until ≈ **25–30% volume loss**; hypotension is late	Falls earlier
Haematuria	May be absent (up to 2/3 of grade ≥ 2 injuries)	Classic trigger
Imaging decision	**Mechanism-led**; keep radiation as low as reasonably achievable (ALARA)	Led by haematuria and hypotension

2 · Indications for imaging

Why this matters: mechanism, not urine, decides who is scanned.

Setting	Indication (American Urological Association [AUA])
Stable blunt	Gross haematuria
Stable blunt	Microscopic haematuria with systolic blood pressure (SBP) < 90 mm Hg
Mechanism / examination	Rapid deceleration; significant flank blow; rib fracture; flank ecchymosis
Penetrating	Any injury to abdomen, flank or lower chest, even without haematuria

- Paediatric triggers: fall > **10 feet (3 m)**, high-speed collision, flank blow; rib, spine, femur or pelvic fracture. All four markers together (haematuria, hypotension, associated injury, mechanism) give > **98% sensitivity**.
- † Paediatric hypotension (1–10 years) is SBP < 70 + (2 × age) mm Hg.

> **KEY POINT**
> Trap: "no haematuria, so no imaging". Haematuria is one of four routes into imaging; in children both adult markers fail, so mechanism drives the decision.

3 · Imaging studies

Why this matters: a single CT phase misses urine leaks.

- Focused assessment with sonography for trauma (FAST) shows free fluid only, not stage. Renal ultrasound (US) has no radiation and detects grade 3–5; European Association of Urology (EAU) Paediatric uses it for screening.
- **Contrast computed tomography (CT)** is preferred. Children often get one phase, so add a **delayed (excretory) phase** if a leak is suspected; it finds urine, not blood.

4 · Grading

Why this matters: grade predicts intervention.

American Association for the Surgery of Trauma (AAST): grades 1–3 are **low** and 4–5 are **high**.

Grade	1989 AAST	2011 revision
1	Contusion or non-expanding subcapsular haematoma	Subcapsular haematoma/contusion
2	Perirenal haematoma; cortical laceration < 1 cm	Same, within Gerota fascia
3	Laceration > 1 cm, no collecting-system injury	Same, into medulla
4	Laceration into collecting system; main artery/vein injury with contained haemorrhage	Collecting system with extravasation; segmental vessel injury; pelvis laceration or **UPJ disruption**; shattered kidney
5	**Shattered kidney**; hilar avulsion	Main artery/vein laceration, avulsion or thrombosis

KEY POINT

Grades 1–3 differ only in depth (1 cm) and never involve the collecting system — entry makes grade 4. Shattered kidney: grade 5 (1989), grade 4 (2011).

High-risk CT criteria (grade 4–5)

Criterion	Predicts
Perirenal haematoma > 3.5 cm adult, > **2.2 cm child**	Bleeding needing angioembolisation
Vascular contrast extravasation; medial laceration	Bleeding
> 25% parenchyma devitalised	Bleeding and open surgery
Medial urinary extravasation	UPJ disruption
Extravasation with no contrast in distal ureter	Urine leak needing intervention

Risk adds up: with 0–1 factor, < 10% need intervention; with 2–3, > 50% do (grade 4a vs 4b).

5 · Non-operative management (NOM)

Why this matters: observation saves the most kidneys.

- NOM is the **default at every grade**, including grade 4–5 and stable penetrating injury, provided the child is stable (after resuscitation if needed).

ALGORITHM

1. Confirm stability; grade the injury.
2. Bed rest, with serial haemoglobin and abdominal examination.
3. When gross haematuria resolves and counts are stable, mobilise.

- Grade 1–2 needs only **23-hour observation**. Repeat imaging: AUA CT at 48 h (grade 4–5), EAU US at 2–3 days; but **symptoms, not imaging, predict complications**.
- Grade 3 (devitalised) to 5: ≈ **25%** need embolisation, ≈ **15%** urinoma intervention, ≈ 5% exploration.

6 · Complications during NOM

Why this matters: watch for leaks and late bleeds.

- **Urine leaks** occur only in grade 4–5, and up to **85%** resolve. A urinoma shows the triad of **flank pain, ileus and low-grade fever**.

- If no contrast reaches the distal ureter, do cystoscopy and retrograde pyelography (RPG). Escalate from **stent and Foley catheter** (AUA first line, ≈ 90%; out at 4–6 weeks) to percutaneous drain, then nephrostomy, then open repair.

Bleeding	Persistent	Delayed
Mechanism	Vessels fail to tamponade	Fistula or pseudoaneurysm ruptures as the haematoma liquefies
Timing	From injury	Typically **1–2 weeks** (up to 1 month)
Stable	Allow 2–3 transfusions, then embolise	Early embolisation unless mild
Unstable	Urgent angioembolisation	

- Superselective embolisation succeeds in > 80%. Retained function is ≈ 35–45% for grade 4 and ≈ 10–30% for grade 5.

7 · Operative management

Why this matters: surgery is for instability; confirm the other kidney first.

- Operate for **instability from renal bleeding**, pulsatile or expanding haematoma, or failed embolisation. A stable non-pulsatile haematoma found at laparotomy is **left alone**, even after gunshot wounds.
- A **one-shot pyelogram** (2 mL/kg, film at 10–15 min) confirms only a **contralateral kidney**.
- P2 Renal trauma (renal parenchymal and vascular injuries, including grade IV): diagnosis and management — *vascular control, renorrhaphy, nephrectomy; revascularise only solitary/bilateral; Page kidney*

Not in the paper answer:

- Early vascular control did not lower nephrectomy in the one randomised trial.
- Post-embolisation syndrome (pain, fever, ileus) settles in 3–4 days; persistent fever needs CT for abscess.

ALGORITHM

1. Confirm a contralateral kidney.
2. Open Gerota fascia; control the vessels; debride; close the collecting system, then parenchyma over a bolster.
3. Drain. If the injury is penetrating, inspect the ureter.

- **UPJ disruption** follows deceleration with hyperextension, with †**Chance (lap-belt) fracture**, aortic and duodenal injury; haematuria may be absent.
- CT shows **medial extravasation, empty distal ureter and often no laceration**; confirm with RPG. Stable and early: stented ureteropelvic anastomosis. Unstable or > 1 week: nephrostomy, repair at 2–3 months.
- Pre-existing UPJ obstruction needs an earlier stent, then pyeloplasty at 8–12 weeks. In a duplex kidney, an upper-pole stent will not drain a lower-pole leak.

8 · Follow-up

Why this matters: scarring and hypertension appear late.

- Scarring: grade 1–2 none, grade 3 ≈ 50–60%, grade 4–5 ≈ 100%, so do **US at 2–4 months**. Use a dimercaptosuccinic acid (DMSA) scan if function loss is suspected.

- **Hypertension** (> 30 days) is rare and renin-mediated, so use renin–angiotensin blockade (angiotensin-converting enzyme inhibitor or receptor blocker). Check blood pressure yearly after grade 3–5. Solitary kidney: sport with padding, but no motorised sport.

9 · Ureteral trauma

Why this matters: ureteric injury is easily missed.

- Most are penetrating. **Haematuria is absent in 25–50%** and ≈ 10% are missed at laparotomy, so diagnose by **delayed-phase CT** or RPG.

AAST grade	Injury
I	Contusion/haematoma, no devascularisation
II / III	< 50% / ≥ **50%** transection
IV	Complete transection, < 2 cm devascularisation
V	Avulsion, > **2 cm** devascularisation

- Found within 1 week: repair if stable; ligate and place a nephrostomy if unstable. Found after 5–7 days: stent first; if that fails, nephrostomy and reconstruction at 6–12 weeks.
- P3 Ureteric injury: iatrogenic (gynaecological) and upper ureteric injury — *repair by level: ureteroureterostomy, Boari, reimplant ± psoas hitch; ileal ureter, autotransplant*

Not in the paper answer:

- **Avoid transureteroureterostomy in children**, because a long life puts the other kidney at risk.
- After high-velocity gunshot, a contused ureter may necrose days later, so stent and drain if in doubt. Missed injury means more nephrectomy.
- Follow-up: US 4–6 weeks after stent removal, then yearly for 2–3 years; hydronephrosis needs a mercaptoacetyltriglycine (MAG3) diuretic renogram.

10 · Bladder injury

Why this matters: the child's bladder sits higher, so injury patterns differ.

Feature	Child	Adult
Bladder position	**More intraperitoneal**, so isolated injury is commoner (≈ half have no fracture)	Pelvic organ
Pelvic fracture	Half as frequent, so it implies higher energy and more urethral injury	Commoner
Posterior urethral injury	Prostatic; **bladder neck injury commoner**	Bulbomembranous

- **Gross haematuria or retention** calls for **retrograde cystography** with retrograde urethrography (RUG); delayed-phase CT may miss a rupture.
- AAST: I contusion; II extraperitoneal < 2 cm; III extraperitoneal ≥ 2 cm or intraperitoneal < 2 cm; IV intraperitoneal ≥ 2 cm; V bladder neck or trigone.
- Extraperitoneal tears get a catheter, but are repaired for bone fragments, hardware, rectal or vaginal tear, or bladder neck injury. Intraperitoneal tears are **always repaired**. Keep the catheter 1–2 weeks, then do a cystogram.

KEY POINT

Bladder neck avulsion is commoner in children; girls depend more on the bladder neck for continence, so outcome is poor — many need secondary procedures, even closure and diversion.

11 · Urethral injury

Why this matters: image before any catheter.

Grade/type	AAST	Goldman
I	Contusion: meatal blood, normal RUG	Posterior urethra stretched, intact
II	Stretch, no extravasation	Rupture above intact urogenital diaphragm
III	Partial: extravasation, contrast reaches bladder	Rupture through diaphragm into bulbar urethra
IV	Complete, no bladder filling, < 2 cm gap	Bladder rupture extending into proximal urethra
V	≥ 2 cm gap or into prostate/vagina	Anterior urethral injury

- Do **RUG before any catheter**. If a catheter passes, the tear is partial, and its stricture recurs in half after urethrotomy. If not, place a **suprapubic tube**.
- Realignment is a brief attempt only; EAU advises against immediate urethroplasty. **Bladder neck injury is repaired at once.**

ALGORITHM

1. Divert with a suprapubic tube and wait several months.
2. Assess with cystogram and RUG, with or without endoscopy.
3. Do transperineal excision and primary anastomosis; if there is tension, add pubectomy.
4. Catheter for 2–3 weeks; cystogram; remove the suprapubic tube.

- Restricture ≈ 10%, incontinence ≤ 20%, erectile dysfunction up to half (gap > 2.5 cm). Girls: ≈ 5% of pelvic fractures, with vaginal or rectal injury.

12 · Genital injury

Why this matters: always consider abuse.

KEY POINT

Most are accidental, but **abuse must always be assessed** (≈ 1 in 7 boys, 1 in 3 girls abused). Examine anus and rectum; involve social services before discharge.

- Penis AAST: I skin; II Buck fascia; III avulsion, glans/meatus laceration, defect < 2 cm; IV partial penectomy or ≥ 2 cm; V total penectomy.
- **Circumcision** is the commonest cause; the **Mogen clamp** causes the worst (glans amputation). Lyse adhesions carefully (see Ch 55). A **hair tourniquet** is painless, so it may present late with fistula or amputation.
- Testis AAST: I contusion; II subclinical tunica tear; III < 50% loss; IV ≥ 50% loss; V destruction. If US shows tunica discontinuity, haematocele or focal flow loss, **explore and close the tunica**. Repeat US at 6 weeks to exclude a masked tumour.
- Vulval injury is from straddle in ≈ 80% and is labial. Hymenal, vaginal or perianal injury means **penetration: evaluate for abuse**; a **normal examination does not exclude abuse**.

★ Putting it together

A · Renal colic

Why this matters: first decide whether to drain.

ALGORITHM

1. Give a non-steroidal anti-inflammatory drug (NSAID; ketorolac 30 mg) unless there is renal failure, bleeding or pregnancy. Non-contrast computed tomography (NCCT) confirms; use ultrasound first in pregnancy and children (see Ch 96).
2. If there is infection, bilateral obstruction, a solitary kidney, immunosuppression, renal deterioration or refractory pain or vomiting, then **decompress now** (stent or nephrostomy; equal). Never lithotripsy while infected.
3. **Otherwise**, if the stone is <10 mm, try passage with or without an α-blocker (distal 5–10 mm) for 4–6 weeks. If that fails, do ureteroscopy (URS) or shock wave lithotripsy (SWL) (see Ch 97).

B · Stone operation

Why this matters: size and site pick the operation.

Stone	First choice	Notes
Ureteric, any level	URS or SWL	URS clears more; SWL has fewer complications. Hard proximal >15–20 mm: consider percutaneous nephrolithotomy (PCNL)
Renal <2 cm, not lower pole	SWL or URS	SWL works best <900 Hounsfield units and skin-to-stone <10 cm
Lower pole >1 cm	PCNL or URS	SWL not first line (Lower Pole I)
Burden >2 cm, staghorn	PCNL	Staged URS an option

KEY POINT

SWL absolute contraindications: pregnancy, coagulopathy or antithrombotics, untreated infection, uncontrolled hypertension, arterial aneurysm, and obstruction distal to the stone. No routine stent after SWL or uncomplicated URS (see Ch 98).

C · Prevention

Why this matters: advice for all, tests for the high-risk.

ALGORITHM

1. **All patients:** urine ≥2.5 L/day †, normal calcium, low sodium and less animal protein (see Ch 99).
2. **If high risk**, collect 24-hour urines, give targeted therapy and repeat the urines within 6 months.

Abnormality	Treatment
Hypercalciuria	Thiazide and potassium citrate (hypokalaemia lowers citrate)
Hypocitraturia	Potassium citrate

Abnormality	Treatment
Uric acid	Alkalinise to pH 6.5–7 †; allopurinol only for hyperuricosuria/gout or failed alkali
Cystine	≥3 L/day fluid, pH 7.0–7.5, then tiopronin
Struvite	Complete clearance; check metabolism

D · Renal trauma

Why this matters: most stable kidneys heal without surgery.

ALGORITHM

1. **Image** with delayed-phase CT for gross haematuria, microhaematuria with shock, deceleration or penetrating injury (see Ch 94).
2. If stable with grade I–IV (or selected V), manage **non-operatively** (>90% succeed). If there is a contrast blush, embolise; if a leak is symptomatic, stent.
3. If unstable or the haematoma is expanding, **explore**; a one-shot pyelogram (2 mL/kg, 10 min) confirms the other kidney.

E · Bladder and urethra

Why this matters: repair now or deliberately later.

Injury	Management
Extraperitoneal bladder rupture	22 Fr catheter, cystogram at 2–3 weeks; repair if at laparotomy, bone spicule, or unhealed at 4 weeks
Intraperitoneal bladder rupture	† Surgical repair
Pelvic fracture urethral injury	Suprapubic tube (or brief realignment) now, delayed urethroplasty later
Penile fracture	Prompt tunical repair; check the urethra (see Ch 125)
Testicular rupture	Immediate repair

Cystography must be retrograde, filled to ≥350 mL.

F · Ureteric defects

Why this matters: defect length chooses the repair.

Defect	Repair
2–3 cm, upper/mid	Spatulated ureteroureterostomy
Lower ureter, 6–10 cm	Psoas hitch reimplant
10–15 cm	Boari flap
Longer	Transureteroureterostomy, ileal ureter (creatinine <2 mg/dL), autotransplant, nephrectomy (see Ch 93)

G · Children

Why this matters: children differ.

KEY POINT

Stones: ultrasound first, and a full metabolic evaluation after the first stone. Renal ≤20 mm: SWL or URS; >20 mm: PCNL (see Ch 54). **Trauma:** image by mechanism (fall >3 m, deceleration), because blood pressure falls late; a perirenal haematoma >2.2 cm predicts embolisation (see Ch 62).